

Untitled hardware project

Class bioreactor	Feasibility conditional	BoM Readiness Ready	Score 7.7
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Executive Summary

Metric	Value	Unit	Confidence	Notes
Annual culture volume	2000	L/year	medium	Working volume multiplied by assumed annual batch count until process recipe is known.
Estimated CAPEX	0	GBP	low	No CAPEX claim admitted yet; supplier-backed BoM evidence is required before costs appear.
Estimated annual OPEX	0	GBP/year	low	No OPEX claim admitted yet because CAPEX has no sourced BoM basis.

Brief Intake Gate

brief_intake_review_required

5/6 intake areas pass; 1 review, 0 blocked. Class bioreactor (high); requirements 1; numeric facts 1; requirement trace 1/1 covered; scratch grammar yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_text	pass	1	231 non-whitespace character(s) in the source brief.	none	Brief text is present.
requirement_quantification	review	0.5	1 quantified requirement(s), 1 numeric fact(s).	none	Proceed only as assumption-heavy review; ask for more quantified requirements before treating outputs as stable.
product_class_selection	pass	1	Selected bioreactor with high confidence.	none	Product class selection is strong enough for scratch architecture.
scratch_design_support	pass	1	bioreactor has a deep scratch architecture grammar.	none	Use scratch universal architecture flow for this brief.
requirement_trace_seed	pass	1	1 covered, 0 partial, 0 uncovered requirement trace row(s).	none	Extracted requirements have trace seeds into the architecture.
assumption_boundary	pass	1	1 brief assumption note(s); no-quantified-default path: no.	none	Assumption boundary is explicit enough for first-pass architecture.

Brief Clarification Plan

clarification_recommended

4 clarification question(s): 0 required, 3 recommended, 1 optional. Architecture blockers 0. Next question: target_metrics:1.

#	Kind	Priority	Status	Question	Expected Answer	Example	Blocks Architecture
1	target_metrics	major	recommended	What are the top two or three quantified performance targets?	Metric name, value and unit for each target.	Target output, capacity or throughput; runtime or duty cycle; size, mass or cost constraint.	no
2	operating_envelope	major	recommended	What operating envelope should the design satisfy?	A short list of dimensions, duty cycle, environment, runtime, capacity or output constraints.	Dimensions, deployment environment, runtime/duty cycle, electrical/fluid/data boundaries.	no
3	interfaces_and_integration	major	recommended	What external systems, users or infrastructure must this design interface with?	List each external interface and any protocol, voltage, fluid, data or mechanical constraint.	Power, data, mechanical, thermal, fluid, user and service interfaces.	no
4	compliance_and_risk	minor	optional	Are there named standards, hazards or certification targets that must be treated as in-scope?	Named standard, jurisdiction, hazard or certification target.	Relevant standards, jurisdiction, safety hazards and certification target.	no

Architecture Admission Gate

architecture_generation_review_required

4/6 admission areas pass; 2 review, 0 blocked. Can use for review: yes; can proceed to BoM: no. Source scratch_universal_architecture; next action: Treat generated architecture as review-only until recommended clarification questions are answered..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_intake_clearance	review	0.83	5/6 intake areas pass; class bioreactor (high); 1 quantified requirement(s).	none	Treat generated architecture as review-only until recommended clarification questions are answered.
clarification_clearance	review	1	4 clarification question(s): 0 required, 3 recommended, 0 architecture-blocking.	none	Keep the architecture labelled as review-only until recommended brief clarifications are answered.
class_grammar_clearance	pass	1	Product class bioreactor; scratch universal grammar supported: yes.	none	Route this class through the scratch universal module grammar.
scratch_stage_clearance	pass	1	Architecture stage passed; source scratch_universal_architecture; modules 9; submodules 27; components 108.	none	Architecture was produced by the scratch universal architecture stage.
architecture_readiness_boundary	pass	1	Architecture ready for BoM: yes; blocking issues 0; required links 6/6.	none	Deterministic architecture readiness checks passed; sourcing work may begin only with explicit evidence.
bom_provenance_held	pass	1	0 admitted sourcing evidence row(s), 0 source-backed BoM claim(s), 0 provenance violation(s).	none	BoM fields remain behind the explicit sourcing-evidence boundary.

Stage Integrity Gate

stage_trace_accepted

5/5 stage-integrity areas pass; 0 review, 0 blocked. Stages 8/8; ordered: yes; architecture source scratch_universal_architecture; admitted priced lines 0; provenance violations 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
stage_sequence	pass	1	8/8 expected compiler stages are present in order.	none	Stage sequence matches the scratch compiler contract.
stage_payloads	pass	1	8/8 stages have metrics; 8/8 have evidence or explicit limitation; 8/8 have limitations.	none	Every stage carries metrics, evidence or explicit limitation, and limitations.
scratch_architecture_source	pass	1	Product class bioreactor; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Architecture came from the scratch universal grammar.
architecture_before_sourcing	pass	1	Architecture ready: yes; readiness stage passed; sourcing stage warning.	none	Sourcing stage is downstream of architecture readiness and has a consistent status.
sourcing_provenance_boundary	pass	1	0 admitted source evidence row(s), 0 admitted priced line(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN and cost fields stay behind admitted sourcing evidence.

Scratch Lineage Gate

scratch_lineage_clean

6/6 lineage areas pass; 0 review, 0 blocked. Architecture source scratch_universal_architecture; design refs 236; source refs 0; forbidden refs 0; forbidden stage mentions 0; chain-v2 benchmark no (prototype internal depth target).

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_source	pass	1	Product class bioreactor; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Design content came from the scratch universal architecture stage.
design_provenance_refs	pass	1	460 provenance ref(s); 236 design/architecture ref(s); 0 invalid kind(s); 0 forbidden ref(s).	none	All design provenance references are in the allowed scratch provenance vocabulary.
bom_candidate_provenance	pass	1	108 candidate BoM line(s); 0 admitted source evidence row(s); 0 unprovenanced supplier/manufactur er/MPN/cost claim(s).	none	Candidate BoM fields remain unpriced or explicitly source-backed.
source_evidence_boundary	pass	1	0 source provenance ref(s); 0 outside admitted evidence boundary; 0 inside architecture content.	none	Source refs appear only in admitted source evidence, source ledger entries or matching source-backed BoM lines.
chain_v2_quarantine	pass	1	0 forbidden provenance ref(s); 0 forbidden stage-trace mention(s).	none	No chain-v2/state.json/modifi er lineage appears inside generated design content or stage traces.
benchmark_isolation	pass	1	Benchmark source: prototype internal depth target; chain benchmark used: no; policy: Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.	none	Benchmark does not use chain-v2 input.

Architecture Freeze Gate

architecture_freeze_review_required

1/6 freeze areas pass; 5 review, 0 blocked. Modules/submodules/components 9/27/108. Structurally ready for sourcing yes; independent review accepted no. Module engineering module_engineering_review_required; interfaces interface_review_ready; requirements coverage_review_ready; next action: Keep architecture labelled review-ready until admission, stage trace and scratch lineage are all clean..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
admission_and_lineage	review	0.88	Architecture admission architecture_generation_review_required; stage integrity stage_trace_accepted; scratch lineage scratch_lineage_clean.	none	Keep architecture labelled review-ready until admission, stage trace and scratch lineage are all clean.
module_structure	review	0.75	Component allocation allocation_ready; module engineering module_engineering_review_required; 0/9 modules pass; 6 modules still have critical sourcing blocks.	none	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.
interface_contracts	review	1	Interface verification interface_review_ready; 6/6 required interfaces have carriers; 0 accepted.	none	Collect or accept interface-review evidence before calling the architecture fully frozen.
requirement_coverage	review	1	Requirement coverage coverage_review_ready; 1/1 requirements structurally ready or accepted; accepted ratio 0.	none	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.
engineering_review_state	review	0.67	Pre-BoM engineering engineering_review_ready; 4/6 areas pass; calculations needing review 2; assumptions needing review 21.	none	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.
sourcing_boundary	pass	1	108 candidate BoM lines; 0 admitted sourcing evidence rows; 0 provenance violation(s); 44 critical missing source claim(s).	none	Sourcing and BoM claims remain behind explicit admitted source evidence.

Architecture Freeze Closure Plan

architecture_freeze_review_required

5 closure row(s): 5 ready, 0 blocked. Queues S/E/V/R 1/1/2/1; next row freeze-close:1:admission_and_lineage. Structurally ready for sourcing yes; independent review accepted no.

#	Area	Queue	Priority	Status	Required Evidence	Acceptance Criteria	Input Artifacts
1	admission_and_lineage	architecture_review	minor	ready	Keep architecture labelled review-ready until admission, stage trace and scratch lineage are all clean.	Architecture admission verdict is architecture_generation_admitted. Stage integrity verdict is stage_trace_accepted. Scratch lineage verdict is scratch_lineage_clean.	*.architecture-admission-gate.csv; *.stage-integrity-gate.csv; *.scratch-lineage-gate.csv
2	module_structure	sourcing_intake	major	ready	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.	Module engineering verdict is module_engineering_ready. No module has blocked allocation or unresolved critical sourcing blockers. Every module review question is accepted, revised or explicitly deferred.	*.module-engineering-gate.csv; *.component-allocation-gate.csv; *.engineering-review-pack.csv; *.sourcing-intake-template.csv
3	interface_contracts	verification_intake	minor	ready	Collect or accept interface-review evidence before calling the architecture fully frozen.	Interface verification verdict is accepted_interfaces. Every required interface has endpoint carriers and accepted reviewer evidence.	*.interface-contracts.json; *.interface-verification-gate.csv; *.verification-intake-template.csv
4	requirement_coverage	verification_intake	minor	ready	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.	Requirement coverage verdict is accepted_evidence. Every parsed requirement has architecture links and accepted verification evidence.	*.requirement-coverage-gate.csv; *.engineering-assurance-matrix.csv; *.verification-intake-template.csv

#	Area	Queue	Priority	Status	Required Evidence	Acceptance Criteria	Input Artifacts
5	engineering_review_state	engineering_review	major	ready	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.	Pre-BoM engineering verdict is engineering_accepted. Calculations, assumptions and verification activities are accepted or explicitly deferred.	*.pre-bom-engineering-gate.csv; *.engineering-calculations.csv; *.engineering-assumptions.csv; *.engineering-verification-plan.csv

Component Candidate Gate

component_candidates_ready_for_sourcing

7/7 candidate areas pass; 0 review, 0 blocked. Candidate lines 108; critical 11/11; worklist rows 108; duplicate groups 0; provenance violations 0; ready for sourcing yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
candidate_presence	pass	1	108 candidate BoM line(s), 108 distinct component identity key(s).	none	Component candidates exist for downstream sourcing review.
candidate_identity	pass	1	108/108 candidate line(s) have non-generic identity, description and role context.	none	Candidate identities are specific enough for first-pass sourcing worklists.
quantity_basis	pass	1	108/108 candidate line(s) have positive quantity and unit.	none	Candidate quantities have positive values and explicit units.
critical_candidate_coverage	pass	1	11/11 class-critical part(s) are represented by candidate lines; 11 critical BoM line(s).	none	Class-critical parts have candidate lines before sourcing begins.
duplicate_identity_review	pass	1	0 duplicate component identity group(s), 0 involving critical lines.	none	No duplicate component identities need sourcing review.
sourcing_worklist_alignment	pass	1	108/108 unpriced candidate line(s) appear in the sourcing worklist.	none	Every unpriced candidate line is visible in the sourcing worklist.
provenance_boundary	pass	1	0 source evidence row(s), 0 source-backed claim(s), 0 provenance violation(s).	none	Candidate sourcing fields remain clean unless admitted evidence supplies them.

Sourcing Authorization Gate

sourcing_authorization_review_required

5/6 sourcing-authorization areas pass; 1 review, 0 blocked. Critical intake 11/11; full intake 108/108; admitted evidence 0; rejected evidence 0; provenance violations 0; authorized no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_authorized	review	0.5	architecture_generation_review_required; review yes, BoM no.	none	Keep sourcing as review-only until architecture admission is fully cleared.
component_candidate_authorization	pass	1	component_candidates_ready_for_sourcing; 108 candidate line(s), 108 sourcing-worklist row(s), 0 provenance violation(s).	none	Component candidates are concrete enough for sourcing intake.
critical_intake_scope	pass	1	11/11 critical unpriced line(s) appear in the critical sourcing intake template.	none	Critical sourcing intake scope is complete.
full_worklist_scope	pass	1	108/108 total unpriced line(s) appear in the full sourcing intake template.	none	Full sourcing intake scope matches the unpriced worklist.
evidence_admission_boundary	pass	1	0 admitted source evidence row(s), 0 rejected source evidence row(s).	none	No rejected sourcing evidence is waiting for repair.
provenance_boundary	pass	1	0 source-backed BoM claim(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN, lead-time and cost fields remain behind admitted evidence.

BoM Admission Gate

candidate_bom_review_required

Display mode candidate_only. 1/7 BoM-admission areas pass; 6 review, 0 blocked. Critical priced 0/11; source-backed claims 0; provenance violations 0. Candidate BoM yes; priced BoM no; procurement use no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_authorization	review	0.83	sourcing_authorization_review_required ; authorized no; 5/6 areas pass.	none	Keep BoM admission review-only until sourcing authorization clears.
candidate_bom_scope	pass	1	108 candidate BoM line(s), 11 critical line(s).	none	Candidate BoM scope exists for sourcing and evidence collection.
source_admission_state	review	0	0 admitted source row(s), 0 rejected source row(s), 0/108 priced line(s).	none	No source rows are admitted; keep this as a candidate-only BoM.
critical_bom_admission	review	0	0/11 critical line(s) have admitted unit costs; 11 critical line(s) remain unpriced.	none	Start with critical sourcing intake before treating any BoM cost as admitted.
source_reference_quality	review	0	no_sourcing_evidence; 0/0 source-quality rows pass; 0 protocol fixture row(s).	none	No source evidence is present; keep the BoM candidate-only.
provenance_boundary	review	0.92	0 source-backed claim(s), 44 critical missing-source claim(s), 0 provenance violation(s).	none	Missing critical source claims are acceptable only in candidate-only mode; collect evidence before priced BoM use.
display_mode	review	0.25	Display mode will be candidate_only.	none	Show candidate BoM as a sourcing queue with supplier, MPN and cost fields empty.

Report Readiness Gate

architecture_review_ready

5/6 sections at or above 8. Mean score 7.67. Architecture ready: yes. Critical unpriced lines: 11. Verification accepted: 0/26; unaccepted: 26. Verification acceptance ratio: 0.

- bom score 1 is below 8.
- 11 critical BoM line(s) lack source-backed cost evidence.
- 6 verification activity/activities remain blocked.
- 0/26 non-sourcing verification activity/activities have accepted reviewer evidence.

Section	Score	Target	B/M/m Issues	Action
executive_summary	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
brief_requirements	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
design_modules	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
bom	1	below target	11/1/0	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates.
feasibility_notes	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
sources_references	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.

Pre-BoM Engineering Gate

engineering_review_ready

4/6 areas pass; 2 review, 0 blocked. Modules/submodules/components: 9/27/108. Calculations within envelope 0/2; calculation blockers 0. Assumption review rows 21; architecture blockers 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	9 modules, 27 submodules and 108 component words pass architecture readiness.	none	Architecture is structurally ready for engineering review.
component_allocation	pass	1	27/27 submodules ready; 11/11 critical parts allocated; 0 duplicate component groups.	none	Component candidates are allocated well enough for engineering review.
interface_verification	pass	1	6/6 required interfaces have endpoint carriers; 0 accepted with evidence.	none	Required interfaces are structurally ready for reviewer evidence.
requirement_coverage	pass	1	1/1 requirements are structurally ready or accepted; accepted evidence ratio 0.	none	Requirements are structurally ready for reviewer evidence.
calculation_envelope	review	0	0/2 calculations are within envelope; 2 need review; 0 block.	none	Attach reviewer evidence to calculations marked needs_review.
engineering_assumptions	review	0.25	7/28 assumptions are brief-supported or model-present; 21 need review/source evidence; 0 block architecture.	none	Review assumptions and keep BoM/source-required assumptions out of trusted claims until evidence is admitted.

BoM Costing Gate

costing_not_started

1/7 costing areas pass; 0 review, 6 blocked. Critical priced lines 0/11; unpriced critical lines 11. Source-backed claims 0; provenance violations 0; sourcing evidence 0/0 production-ready, 0 protocol. Source quality: no_sourcing_evidence (0/0 pass, 0 placeholder URL, 0 quote-anchored). BoM total £0; CAPEX £0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_admission	blocked	0	0/108 BoM lines have admitted pricing; rejected source records 0.	structure_containment_single_use_bag_chamber_single_use_bioreactor_bag: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_sterile_connector_panel_sterile_tubing_and_connector_set: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. actuation_kinematics_agitation_drive_train_agitation_drive: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. mass_fluid_transport_process_feed_pump_manifold_peristaltic_feed_pump: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. mass_fluid_transport_process_sparger_gas_mixing_sparger_and_gas_mix_manifold: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 6 more	Admit source-backed supplier, manufacturer, MPN and unit-cost evidence before costing review.
critical_line_pricing	blocked	0	0/11 critical lines have admitted unit costs.	structure_containment_single_use_bag_chamber_single_use_bioreactor_bag: Single-use bioreactor bag structure_containment_sterile_connector_panel_sterile_tubing_and_connector_set: Sterile tubing and connector set actuation_kinematics_agitation_drive_train_agitation_drive: Agitation drive mass_fluid_transport	Price every critical line from admitted source evidence.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				rt_process_feed_pump_manifold_peristaltic_feed_pump: Peristaltic feed pump mass_fluid_transport_process_sparger_gas_mixing_sparger_and_gas_mix_manifold: Sparger and gas-mix manifold plus 6 more	
source_backed_provenance	blocked	0.92	0 source-backed claims; 44 critical missing-source claims; 0 provenance violations.	structure_containment_single_use_Bag_chamber_single_use_bioreactor_bag/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_single_use_Bag_chamber_single_use_bioreactor_bag/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_single_use_Bag_chamber_single_use_bioreactor_bag/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_single_use_Bag_chamber_single_use_bioreactor_bag/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_sterile_connector_panel_sterile_tubing_and_connector_set/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before	Keep supplier, manufacturer, MPN, unit cost and lead-time fields empty until source-backed evidence is admitted.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				BoM can pass. plus 39 more	
component_identity	pass	1	0 duplicate component identity group(s); 0 include critical lines.	none	Component identities are unique enough for line-level costing.
sourcing_evidence_authenticity	blocked	0	0/0 sourcing evidence rows are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Add sourcing evidence rows before BoM costing can start.
source_reference_quality	blocked	0	0/0 sourcing refs pass quality checks; 0 placeholder URL, 0 quote-anchored, 0 fresh timestamp.	none	Add source references before costing review.
cost_totals	blocked	0	BoM total 0 GBP; CAPEX 0 GBP; annual OPEX 0 GBP.	11 critical line(s) are unpriced, so totals are incomplete.	Do not use cost totals until every critical line has an admitted unit cost.

Evidence Acquisition Plan

42 missing-evidence row(s): 14 sourcing, 28 verification. Ready 39; blocked 0. Blocks BoM 14; blocks publish 42. Next row: acquire:4:assumption:metric:capex_gbp.

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
1	sourcing	blocker	open	readiness:2	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
2	verification	blocker	open	readiness:3	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
3	verification	blocker	open	readiness:4	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
4	sourcing	blocker	ready_for_intake	assumption:metric:capex_gbp	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						sourceGrade; evidence.ref; evidence.quote; retrievedAt		
5	sourcing	blocker	ready_for_intake	assumption:metric:opex_annual_gbp	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
6	sourcing	blocker	ready_for_intake	single_use_bioreactor_bag	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
7	sourcing	blocker	ready_for_intake	sterile_tubing_and_connector_set	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.ref ; evidence.quote; retrievedAt		
8	sourcing	blocker	ready_for_in take	agitation_drive	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
9	sourcing	blocker	ready_for_in take	peristaltic_feed_pump	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
10	sourcing	blocker	ready_for_in take	sparger_and_gas_mix_manifold	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.quote; retrievedAt		
11	sourcing	blocker	ready_for_intake	temperature_control_loop	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
12	sourcing	blocker	ready_for_intake	exhaust_filter_and_pressure_relief_set	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
13	sourcing	blocker	ready_for_intake	dissolved_oxygen_optical_sensor	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						ote; retrievedAt		
14	sourcing	blocker	ready_for_in take	single_use_ ph_sensor	supplier_cat atalogue_or_q uote	component WordId; supplierNa me; manufactur er; mpn where applicable; unitCostGb p; leadTimeWe eks where stated; sourceGrad e; evidence.ref ; evidence.qu ote; retrievedAt	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes
15	sourcing	blocker	ready_for_in take	load_cell_se t	supplier_cat atalogue_or_q uote	component WordId; supplierNa me; manufactur er; mpn where applicable; unitCostGb p; leadTimeWe eks where stated; sourceGrad e; evidence.ref ; evidence.qu ote; retrievedAt	LLM- estimated cost; benchmark average without supplier evidence; class-pack default; test- fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
16	sourcing	blocker	ready_for_intake	bioreactor_controller	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture://source	yes
17	verification	major	ready_for_intake	design_review:structure_containment	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture://review; source-evidence claims that belong in sourcing intake	no
18	verification	major	ready_for_intake	calculation:structure_containment	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture://review; source-evidence claims that belong in sourcing intake	no
19	verification	major	ready_for_intake	design_review:actuation_kinematics	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture://review; source-evidence claims that belong in	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
							sourcing intake	
20	verification	major	ready_for_intake	calculation: actuation_kinematics	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
21	verification	major	ready_for_intake	design_review:mass_fluid_transport_process	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
22	verification	major	ready_for_intake	calculation: mass_fluid_transport_process	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
23	verification	major	ready_for_intake	design_review:environmental_interface	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; ; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
24	verification	major	ready_for_intake	calculation:environmental_interface	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
25	verification	major	ready_for_intake	design_review:sensing_instrumentation	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
26	verification	major	ready_for_intake	design_review:control_compute_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
27	verification	major	ready_for_intake	design_review:power_distribution	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
28	verification	major	ready_for_intake	design_review:safety_protection	named_engineering_review	activityId; evidenceKind; reviewerNa	anonymous reviewer; unchecked model	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						me; verdict; evidenceRef; evidenceNote; reviewedAt	assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	
29	verification	major	ready_for_intake	design_review:maintenance_serviceability	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
30	verification	major	ready_for_intake	interface_review:structure_containment:mass_fluid_transport_process:sterile_fluid_path	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
31	verification	major	ready_for_intake	interface_review:actuation_kinematics:structure_containment:sterile_boundary	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
32	verification	major	ready_for_intake	interface_review:mass_fluid_transport_process:environmental_interface:gas_path	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNo	anonymous reviewer; unchecked model assertion; test- fixture:// review; source-	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						te; reviewedAt	evidence claims that belong in sourcing intake	
33	verification	major	ready_for_in take	interface_re view:sensin g_instrume ntation:cont rol_comput e_communi cation:sens or_bus	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
34	verification	major	ready_for_in take	interface_re view:control _compute_c ommunicati on:mass_flu id_transport _process:pu mp_control	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
35	verification	major	ready_for_in take	interface_re view:safety_ protection:s tructure_co ntainment:p ressure_reli ef	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
36	verification	major	ready_for_intake	compliance_review:ISO-13485	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
37	verification	major	ready_for_intake	compliance_review:USP-CLASS-VI	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
38	verification	major	ready_for_intake	compliance_review:IEC-61010	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
39	verification	major	ready_for_intake	compliance_review:EU-GMP-ANNEX-11	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
40	verification	major	ready_for_intake	risk_review:1	named_engineering_review	activityId; evidenceKind; reviewerNa	anonymous reviewer; unchecked model	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						me; verdict; evidenceRef; evidenceNote; reviewedAt	assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	
41	verification	major	ready_for_intake	risk_review: 2	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
42	verification	major	ready_for_intake	risk_review: 3	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

Document Trust Gate

evidence_blocked

1/7 trust areas pass; 3 review, 3 blocked. Architecture ready: yes. Claim gate: evidence_blocked. Reviewer evidence 0/26. Source-backed BoM claims 0; critical missing BoM claims 44; provenance violations 0. Evidence authenticity: no_evidence (0/0 production-ready, 0 protocol).

- bom: score 1 below 8
- architecture design has 0 blocked/source-required claim(s) and 36 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s).
- bom sourcing has 55 blocked/source-required claim(s) and 97 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 7 review-required claim(s).
- structure_containment_single_use_bag_chamber_single_use_bioreactor_bag/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_single_use_bag_chamber_single_use_bioreactor_bag/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_single_use_bag_chamber_single_use_bioreactor_bag/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_single_use_bag_chamber_single_use_bioreactor_bag/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_sterile_connector_panel_sterile_tubing_and_connector_set/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- structure_containment_sterile_connector_panel_sterile_tubing_and_connector_set/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

- structure_containment_sterile_connector_panel_sterile_tubing_and_connector_set/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	9 modules, 27 submodules and 6 required links pass deterministic architecture gates.	none	No deterministic architecture blocker remains; continue with reviewer evidence before treating it as accepted.
section_scores	blocked	0.83	5/6 scored sections are at or above 8; mean 7.67.	bom: score 1 below 8	Repair below-target sections through the specific score-ledger actions.
claim_evidence	blocked	0	1/201 claims pass; 57 blocked/source-required, 143 review-required.	architecture design has 0 blocked/source-required claim(s) and 36 review-required claim(s). engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s). bom sourcing has 55 blocked/source-required claim(s) and 97 review-required claim(s). compliance risk has 0 blocked/source-required claim(s) and 7 review-required claim(s).	Accept, correct or reject generated module and submodule allocation claims through engineering review. Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence. Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims. Accept, correct or reject class-pack regulatory and risk claims through compliance/domain review.
requirement_assurance	review	0	0/1 requirements have accepted assurance; 1 need review, 0 blocked, 0 unlinked.	none	Collect reviewer evidence for requirement assurance rows before publication.
reviewer_evidence	review	0	0/26 non-sourcing verification activities accepted; 0 rejected, 0 deferred, 26 pending/blocked.	none	Collect accepted reviewer evidence for every non-sourcing verification activity.
bom_provenance	blocked	0.92	0 source-backed BoM claims, 44 critical missing-source claims, 0 provenance violations.	structure_containment_single_use_bag_chamber_single_use_bioreactor_bag/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_single_use_bag_chamber_single_u	Admit source-backed evidence for critical BoM supplier, manufacturer, MPN and cost claims.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				se_bioreactor_bag/ manufacturer: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. structure_containm ent_single_use_bag _chamber_single_u se_bioreactor_bag/ mpn: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. structure_containm ent_single_use_bag _chamber_single_u se_bioreactor_bag/ unit_cost_gbp: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. structure_containm ent_sterile_connect or_panel_sterile_tu bing_and_connecto r_set/supplier: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. plus 39 more	

evidence_authentici ty	review	0	0/0 evidence row(s) are production- ready; 0 protocol fixture, 0 review/missing metadata.	none	Admit source and reviewer evidence before trusting the document.
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Evidence Authenticity Gate

no_evidence

0/0 evidence rows are production-ready. Protocol fixtures 0; review-required 0; missing metadata 0. External URLs 0; internal references 0; local files 0; unknown references 0.

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
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Evidence Replacement Plan

No evidence replacement rows remain for the current authenticity verdict: no_evidence.

Trust Repair Plan

6 package(s): 6 ready, 0 waiting. Next package: sourcing_intake. Closure rows 53; trust blockers 57. Trust verdict: evidence_blocked.

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
2	Admit Source-Backed BoM Claims	ready	claim_evidence; bom_provenance	14	48	Supplier, manufacturer, MPN, unit cost, lead time, retrieved-at timestamp and source reference for critical BoM lines. Rejected or duplicate component identities resolved before pricing shared items.	*.sourcing-intake-template.csv; *.sourcing-pack.csv; *.bom-provenance-manifest.csv; *.component-identity.csv	BoM provenance has zero critical missing-source claims. Claim Evidence Gate BoM sourcing area is no longer evidence_blocked.
3	Resolve Engineering Review Questions	ready	claim_evidence; requirement_assurance	10	4	Accepted/corrected/rejected answers to review-pack questions. Calculation and assumption decisions tied to reviewer evidence references.	*.engineering-review-pack.csv; *.engineering-assurance-matrix.csv; *.engineering-assumptions.csv; *.engineering-calculations.csv	Requirement assurance rows have accepted or explicitly deferred engineering review evidence. Generated architecture and calculation claims are accepted, corrected or rejected.
4	Admit Reviewer Verification Evidence	ready	reviewer_evidence	28	0	Accepted, rejected or deferred verification evidence for every non-source verification activity.	*.verification-intake-template.csv; *.verification-plan.csv; *.verification-ledger.csv	Every non-source verification activity has accepted reviewer evidence or an explicit defer decision.
5	Validate Evidence Authenticity	ready	evidence_authenticity	0	0	Production supplier/catalogue URLs for BoM source records. Named reviewer evidence references that are not protocol fixtures.	*.evidence-authenticity-gate.csv; *.sourcing-ledger.csv; *.verification-ledger.csv; *.claim-ledger.csv	Evidence Authenticity Gate verdict is production_ready. No protocol_fixture, unknown_reference or missing_metadata evidence

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
								rows remain.
6	Review Compliance And Risk Claims	ready	claim_evidence	0	4	Compliance-domain review of class-pack standards, hazards and mitigation claims.	*.claim-ledger.csv; *.engineering-review-pack.csv	Compliance and risk claims are accepted, corrected or rejected by a domain reviewer.
7	Repair Residual Section Scores	ready	section_scores	1	1	A rerun after evidence intake, plus score-ledger fixes for any section still below target.	*.score-ledger.csv; *.readiness-gate.csv; *.document-trust-gate.csv	All scored sections are at or above target and Document Trust Gate is no longer blocked by scoring.

Claim Ledger

201 claims: 1 brief-supplied, 0 reviewer-accepted, 0 source-backed, 3 calculated needing review, 140 generated needing review, 57 source-required, 0 blocked. Blocking: 57 BoM, 200 publishable.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	Agitation drive / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Agitation drive / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Agitation drive / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Agitation drive / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Bioreactor controller / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Bioreactor controller / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	Bioreactor controller / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Bioreactor controller / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Exhaust filter and pressure relief set / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Exhaust filter and pressure relief set / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Exhaust filter and pressure relief set / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Exhaust filter and pressure relief set / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Temperature control loop / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Temperature control loop / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Temperature control loop / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Temperature control loop / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Peristaltic feed pump / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Peristaltic feed pump / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/ma

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Peristaltic feed pump / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Peristaltic feed pump / unit_cost_g bp	unit_cost_g bp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Sparger and gas-mix manifold / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Sparger and gas-mix manifold / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Sparger and gas-mix manifold / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/ MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Sparger and gas-mix	unit_cost_g bp is	none	yes	yes	Admit source-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
			manifold / unit_cost_g bp	required but not source-backed.				backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Dissolved oxygen optical sensor / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Dissolved oxygen optical sensor / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Dissolved oxygen optical sensor / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Dissolved oxygen optical sensor / unit_cost_g bp	unit_cost_g bp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Single-use pH sensor / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	Single-use pH sensor / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Single-use pH sensor / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Single-use pH sensor / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Load cell set / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Load cell set / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Load cell set / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	Load cell set / unit_cost_g bp	unit_cost_g bp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Single-use bioreactor bag / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Single-use bioreactor bag / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Single-use bioreactor bag / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Single-use bioreactor bag / unit_cost_g bp	unit_cost_g bp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Sterile tubing and connector set / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Sterile tubing and connector set / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Sterile tubing and connector set / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Sterile tubing and connector set / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
component_candidate	source_required	generated_model	Agitation drive	Agitation drive is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Bioreactor controller	Bioreactor controller is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								sourcing intake.
component_candidate	source_required	generated_model	Exhaust filter and pressure relief set	Exhaust filter and pressure relief set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Temperature control loop	Temperature control loop is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Peristaltic feed pump	Peristaltic feed pump is an architecture-derived component candidate with quantity 3 each.	model:scratch_universal_architecture.bioreactor	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Sparger and gas-mix manifold	Sparger and gas-mix manifold is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Dissolved oxygen optical sensor	Dissolved oxygen optical sensor is an architecture	model:scratch_universal_architecture.bioreactor	yes	yes	Admit source-backed supplier, manufactur

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				-derived component candidate with quantity 1 candidate.				er, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Single-use pH sensor	Single-use pH sensor is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Load cell set	Load cell set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Single-use bioreactor bag	Single-use bioreactor bag is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Sterile tubing and connector set	Sterile tubing and connector set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
headline_metric	source_required	assumption	Estimated CAPEX	Estimated CAPEX = 0 GBP.	formula:capex=sum(bom.totalCostGbp); class_pack:bioreactor.class_pack	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
headline_metric	source_required	assumption	Estimated annual OPEX	Estimated annual OPEX = 0 GBP/year.	formula:opex=capex*0.04	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
component_candidate	generated_needs_review	generated_model	drive encoder	drive encoder is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	magnetic coupling hub	magnetic coupling hub is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	motor mounting plate	motor mounting plate is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	drive fault relay	drive fault relay is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	emergency drive stop contactor	emergency drive stop contactor is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	motor current sensor	motor current sensor is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	overspeed interlock input	overspeed interlock input is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	impeller guard insert	impeller guard insert is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	mixing validation coupon	mixing validation coupon is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	rocker angle stop	rocker angle stop is an architecture-derived component candidate with	model:scratch_universal_architecture.bioreactor	no	yes	Review whether this is a unique physical item, shared component or separate

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				quantity 1 candidate.				installation instance.
component_candidate	generated_need_review	generated_model	single-use impeller coupling	single-use impeller coupling is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_need_review	generated_model	audit trail storage	audit trail storage is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_need_review	generated_model	batch historian module	batch historian module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.bioreactor	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Showing first 70 of 201 claims; export CSV for the full claim ledger.

Claim Evidence Gate

evidence_blocked

1/201 claims pass by being brief-supplied, source-backed or reviewer-accepted. 57 blocked/source-required claims, 143 review-required claims. Claim pass ratio 0. Blocking: 57 BoM, 200 publishable.

- architecture design has 0 blocked/source-required claim(s) and 36 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 3 review-required claim(s).
- bom sourcing has 55 blocked/source-required claim(s) and 97 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 7 review-required claim(s).

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
brief_requirements	complete	1	1	0	0	0	0	0	1	No claim-evidence action required for this area.
architecture_design	review	36	0	36	0	0	0	36	0	Accept, correct or reject generated module and submodule allocation claims through engineering review.
engineering_math	blocked	5	0	3	2	2	2	5	0	Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence.
bom_sourcing	blocked	152	0	97	55	55	55	152	0	Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims.
compliance_risk	review	7	0	7	0	0	0	7	0	Accept, correct or reject class-pack regulatory and risk claims through compliance

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
										ce/domain review.

Evidence Gap Register

53 open evidence gaps: 17 blockers, 36 major, 0 minor. Closure paths: 14 sourcing intake, 28 verification intake, 10 engineering review, 0 architecture revision. Blocking rows: architecture 0, BoM 14, publishable 53.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
readiness_blocker	blocker	open	Report readiness gate	bom score 1 is below 8.	score_repair	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	11 critical BoM line(s) lack source-backed cost evidence.	sourcing_intake	bom; publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	6 verification activity/activities remain blocked.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	0/26 non-sourcing verification activity/activities have accepted reviewer evidence.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier,

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							manufacturer, MPN and cost evidence for critical BoM lines.
calculation_review	major	needs review	Bioreactor nominal bag volume	62.5 L nominal bag volume at 80% working fill.	engineering_review	publishable	Confirm bag platform, mixing turnaround, foam allowance, sparging rate and working-volume range.
calculation_review	major	needs review	Bioreactor annual working volume	2000 L/year assumes 40 batches per year.	engineering_review	publishable	Confirm batch duration, turnaround, cleaning/assembly time, hold steps and campaign model.
assumption_review	major	needs review	Annual culture volume	Annual culture volume uses working_volume_l * batches_per_year with medium confidence.	engineering_review	publishable	Review the formula, input values and operating envelope before treating this metric as a performance claim.
assumption_review	blocker	ready for intake	Estimated CAPEX	Estimated CAPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	blocker	ready for intake	Estimated annual OPEX	Estimated annual OPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	major	needs review	Single-Use Working Volume	Single-Use Working Volume currently evaluates as pass against 1 to 2000 L for common single-use mammalian-cell systems; larger volumes need platform-	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				specific bag and mixing evidence.			
assumption_review	major	needs review	Sterile Process Closure	Sterile Process Closure currently evaluates as pass against sterile fluid path, gas path, sensor bus and pump control present.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Aseptic And Pressure Safety	Aseptic And Pressure Safety currently evaluates as pass against sterile boundary, pressure relief, exhaust path, alarm bus and service access present.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	ISO-13485	Quality management for medical-device-adjacent manufacturing systems is relevant to this product class in Global / supplier quality.	engineering_review	publishable	Supplier quality evidence, change control and traceability where applicable.
assumption_review	major	needs review	USP-CLASS-VI	Biocompatibility for polymer contact materials is relevant to this product class in US / global supplier evidence.	engineering_review	publishable	Material certificates for single-use bag and tubing contact surfaces.
assumption_review	major	needs review	IEC-61010	Electrical safety for laboratory control equipment is relevant to this product	engineering_review	publishable	Electrical safety design, protective earth and abnormal operation evidence.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				class in UK/EU/global.			
assumption_review	major	needs review	EU-GMP-ANNEX-11	Computerised systems for GMP records is relevant to this product class in EU / pharma operations.	engineering_review	publishable	Batch record integrity, audit trail and access-control evidence.
sourcing_evidence	blocker	ready for intake	Single-use bioreactor bag	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Sterile tubing and connector set	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Agitation drive	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Peristaltic feed pump	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
sourcing_evidence	blocker	ready for intake	Sparger and gas-mix manifold	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Temperature control loop	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Exhaust filter and pressure relief set	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Dissolved oxygen optical sensor	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
sourcing_evidence	blocker	ready for intake	Single-use pH sensor	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Load cell set	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Bioreactor controller	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
verification_evidence	major	ready for intake	Sterile Structure Containment	Review Sterile Structure Containment architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Sterile Structure Containment	Verify Sterile Structure Containment requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Mixing And Agitation	Review Mixing And Agitation architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
verification_evidence	major	ready for intake	Mixing And Agitation	Verify Mixing And Agitation requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Media Gas And Harvest Transport	Review Media Gas And Harvest Transport architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Media Gas And Harvest Transport	Verify Media Gas And Harvest Transport requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Thermal Gas Exhaust Interface	Review Thermal Gas Exhaust Interface architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Thermal Gas Exhaust Interface	Verify Thermal Gas Exhaust Interface requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Process Sensing Instrumentation	Review Process Sensing Instrumentation architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Control Compute Communication	Review Control Compute Communication architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Power Distribution	Review Power Distribution architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
verification_evidence	major	ready for intake	Aseptic And Pressure Safety	Review Aseptic And Pressure Safety architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Setup Calibration And Service	Review Setup Calibration And Service architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Sterile Structure Containment -> Media Gas And Harvest Transport	Verify sterile_fluid_path interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Mixing And Agitation -> Sterile Structure Containment	Verify sterile_boundary interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Media Gas And Harvest Transport -> Thermal Gas Exhaust Interface	Verify gas_path interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Process Sensing Instrumentation -> Control Compute Communication	Verify sensor_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Control Compute Communication -> Media Gas And Harvest Transport	Verify pump_control interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Aseptic And Pressure Safety -> Sterile Structure Containment	Verify pressure_relief interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
verification_evidence	major	ready for intake	System Compliance	Plan evidence for ISO-13485	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for USP-CLASS-VI	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for IEC-61010	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for EU-GMP-ANNEX-11	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Loss of sterility through connector or bag handling	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Cell culture loss from DO or pH control failure	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Disposable bag overpressure	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Closure Plan

4 active phases. Next phase: sourcing_intake. Rows by phase: architecture 0, sourcing 14, engineering 10, verification 28, score repair 1.

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
2	Admit Source-Backed BoM Evidence	14	14/0/0	bom; publishable	Critical BoM costs, suppliers, manufacturers and MPNs must be source-backed before cost claims are credible.	Every critical BoM line has admitted supplier, manufacturer, MPN, unit cost and source reference. BoM provenance manifest has zero critical missing-source claims.	assumption:metric:capex_gbp; assumption:metric:opex_annual_gbp; readiness:2; sourcing:agitation_drive; sourcing:bioreactor_controller; sourcing:dissolved_oxygen_optical_sensor; sourcing:exhaust_filter_and_pressure_relief_set; sourcing:load_cell_set; plus 6 more
3	Complete Engineering Review	10	0/10/0	publishable	Deterministic arithmetic can expose claims, but engineering judgment must review envelopes, assumptions and operating cases.	All calculation and assumption review rows are accepted, revised or explicitly deferred with reviewer rationale.	assumption:compliance:EU-GMP-ANNEX-11; assumption:compliance:IEC-61010; assumption:compliance:ISO-13485; assumption:compliance:USP-CLASS-VI; assumption:metric:headline_output; assumption:sanitary:bioreactor_aseptic_pressure_safety; assumption:sanitary:bioreactor_process_closure; assumption:sanitary:bioreactor_working_volume; plus 2 more
4	Accept Reviewer Evidence	28	2/26/0	publishable	Reviewer acceptance is tracked separately from deterministic scoring and must close before publication.	Every non-source verification activity is accepted, or explicitly deferred and removed from publication criteria.	readiness:3; readiness:4; verification:calculation_kinematics; verification:calculation_environmental_interface; verification:ca

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
							lculation:mass _fluid_transp ort_process; verification:ca lculation:struc ture_containm ent; verification:co mpliance_revi ew:EU-GMP- ANNEX-11; verification:co mpliance_revi ew:IEC-61010; plus 12 more
5	Repair Remaining Section Scores	1	1/0/0	publishable	After evidence intake, rerun scoring to remove residual deterministic section blockers.	Every scored report section is at or above target and readiness verdict is publishable.	readiness:1

Depth Benchmark

prototype internal depth target. Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.

5 meeting benchmark, 2 below benchmark, average comparable ratio 0.93.

Dimension	Scratch	Benchmark	Ratio	Status	Notes	Action
Functional modules	9 modules	9 modules	1	meets	Core architecture breadth.	Add missing functional modules only when the product class genuinely requires them.
Submodules	27 submodules	27 submodules	1	meets	Submodule count is a rough proxy for engineering decomposition depth.	Expand thin modules with concrete sub-functions and carrier interfaces.
Component candidates	108 components	108 components	1	meets	Component-word count is a rough proxy for BoM review surface, not sourcing quality.	Add justified component candidates where submodules still hide assemblies.
Interface link volume	45 links	36 links	1.25	meets	Compares link volume only; chain-v2 grammar links and scratch interface edges are not semantically identical.	Add interface contracts only where they clarify a real energy, material, signal, load or service path.
Admitted priced-line ratio	0 ratio	0.75 ratio	0	below	Prototype target expects most critical lines to become source-backed before publication.	Use sourcing intake to admit source-backed price/manufacturer/MPN evidence; do not copy benchmark estimates.
Sections scoring >=8	5 sections	6 sections	0.83	below	Readiness gate requires every scored section to reach the target.	Clear section-specific readiness actions, especially BoM evidence.
Verification activities	32 activities	22 activities	1.45	meets	Scratch-only evidence workflow depth; not compared to chain-v2 content.	Keep adding verification activities when new requirements, interfaces or standards are introduced.

Brief Requirements

Design a 50 L single-use mammalian-cell bioreactor with sterile bag, agitation drive, sparger gas manifold, peristaltic feed pumps, dissolved oxygen control, pH sensing, temperature loop, exhaust filter and batch-record controller.

Requirement	Value	Unit
Working volume	50	L

Requirement Traceability

Requirement	Status	Value	Architecture Coverage	Evaluators	Notes
Working volume	covered	50 L	Sterile Structure Containment / Single-use bag chamber; Media Gas And Harvest Transport / Feed pump manifold; Mixing And Agitation / Agitation drive train; Thermal Gas Exhaust Interface / Temperature control loop	metric:headline_out put; sanity:bioreactor_working_volume; sanity:bioreactor_process_closure; sanity:bioreactor_as_eptic_pressure_safety	Working volume is traced through sterile containment, transport, mixing, environmental control and sanity checks. Requirement has architecture coverage plus metric or engineering sanity coverage.

Requirement Coverage Gate

coverage_review_ready

1 requirement row(s): 0 accepted with evidence, 1 coverage-ready, 0 needing review, 0 blocked, 0 unlinked. Structural coverage ratio 1; accepted evidence ratio 0.

Requirement	Status	Present Signals	Missing Signals	Modules	Verification Activities	Required Action
Working volume	review ready	architecture_link ; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	structure_containment; mass_fluid_transport_process; actuation_kinematics; environmental_interface	design_review:structure_containment; calculation:structure_containment; design_review:actuation_kinematics; calculation:actuation_kinematics; design_review:mass_fluid_transport_process; calculation:mass_fluid_transport_process; design_review:environmental_interface; calculation:environmental_interface	Collect accepted reviewer evidence; structural requirement coverage is ready for review.

Engineering Assurance Matrix

1 requirements: 0 accepted, 0 ready, 1 needing review, 0 blocked, 0 unlinked. 1 have calculation links, 1 have review questions, 0 have accepted verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
Working volume	needs review	covered	Sterile Structure Containment; Media Gas And Harvest Transport; Mixing And Agitation; Thermal Gas Exhaust Interface	bioreactor_nominal_ba g_volume_l; bioreactor_annual_working_volume_l	assumption_resolution: component: actuation_kinematics_agitation_drive_train_agitation_drive ; assumption_resolution: component: environmental_interface_exhaust_filter_train_exhaust_filter_and_pressure_relief_set; assumption_resolution: component: environmental_interface_temperature_control_loop_temperature_control_loop; assumption_resolution: component: mass_fluid_transport_process_feed_pump_manifold_peristaltic_feed_pump; assumption_resolution: component: mass_fluid_transport_process_spar ger_gas_mixing_sparger_and_gas_mix_manifold; plus 28 more	pending; pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.

Compiler Stage Trace

Stage	Status	Summary	Metrics	Evidence	Limitations
Brief Parsing	passed	Extracted quantified requirements from the user brief before design generation.	requirement_count: 1; numeric_fact_count : 1; requirement_trace_covered: 1; requirement_trace_partial: 0; requirement_trace_uncovered: 0	working_volume_l: 50 L; working_volume_l: covered	Parsing is deterministic and currently recognises only a small set of hardware requirement patterns.
Product-Class Selection	passed	Selected bioreactor with high confidence from brief keywords.	selected_class: bioreactor; confidence: high; keyword_scores: bioreactor:5	bioreactor: 5	Classifier is keyword based; ambiguous multi-domain briefs need an LLM or richer taxonomy pass.
Universal Module Architecture	passed	Built a 9-module bioreactor architecture from the scratch universal architecture grammar.	module_count: 9; architecture_source : scratch_universal_architecture	Sterile Structure Containment; Mixing And Agitation; Media Gas And Harvest Transport; Thermal Gas Exhaust Interface; Process Sensing Instrumentation; Control Compute Communication; plus 3 more	The scratch grammar is hand-authored and deterministic; it still needs independent engineering review.
Submodule Expansion	passed	Expanded each module into engineering submodules with local purpose, interfaces and component candidates.	submodule_count: 27; average_submodules_per_module: 3	structure_containment.single_use_bag_chamber; structure_containment.support_frame_load_path; structure_containment.sterile_connector_panel; actuation_kinematics.agitation_drive_train; actuation_kinematics.mixing_impeller_interface; actuation_kinematics.agitation_safety_feedback; plus 21 more	Submodule completeness is judged by deterministic coverage checks, not by a physics simulator.
Interface Graph	passed	Checked class-required module-to-module interfaces before allowing BoM review.	required_links: 6; present_links: 6; missing_links: 0; graph_nodes: 36; graph_edges: 72; shared_interface_edges: 39; missing_required_graph_edges: 0	structure_containment -> mass_fluid_transport_process via sterile_fluid_path: present; actuation_kinematics -> structure_containment via	Interfaces prove allocation coverage only; they do not yet validate ratings, thermal margins or dynamic behaviour.

Stage	Status	Summary	Metrics	Evidence	Limitations
				sterile_boundary: present; mass_fluid_transpo rt_process -> environmental_inter face via gas_path: present; sensing_instrument ation -> control_compute_c ommunication via sensor_bus: present; control_compute_c ommunication -> mass_fluid_transpo rt_process via pump_control: present; safety_protection - > structure_containm ent via pressure_relief: present	
Component Candidates	passed	Converted allocated component words into a candidate BoM without supplier prices.	candidate_lines: 108; critical_candidate_lines: 11; admitted_price_lines: 0	single_use_bioreact or_bag: Single-use bioreactor bag; bag_support_tray: bag support tray; port_clamp_set: port clamp set; bag_integrity_test_label: bag integrity test label; stainless_support_f rame: stainless support frame; load_cell_mounting _bracket_set: load cell mounting bracket set; plus 24 more	Showing first 30 of 108 component candidates in the stage trace.
Architecture Readiness Gate	passed	Architecture validators found no blocker or major issue, so candidate BoM review can start.	ready_for_bom: true; blocking_issue_count: 0; sanity_pass: 3; sanity_warn: 0; sanity_fail: 0	pass Single-Use Working Volume: 50 L; pass Sterile Process Closure: present; pass Aseptic And Pressure Safety: present	Passing this gate means deterministic architecture checks passed; it is not a certified engineering sign-off.
Sourcing And BoM Admission	warning	No supplier, manufacturer, part number or cost claims have been admitted because no source-backed evidence was provided.	admission_status: not_started; candidate_lines: 108; admitted_lines: 0; unpriced_lines: 108; unpriced_critical_lines: 11; rejected_source_records: 0	none	BoM cost remains intentionally blocked until 11 critical candidate line(s) have explicit source evidence.

Engineering Architecture Review

Ready for BoM review

9 modules, 27 sub-modules, 108 allocated component words.

From	To	Interface	Status	Engineering Reason
structure_containment	mass_fluid_transport_process	sterile_fluid_path	OK	The sterile bag must connect to feed, harvest and sampling fluid paths.
actuation_kinematics	structure_containment	sterile_boundary	OK	Mixing drive must couple into the disposable culture boundary without breaking sterility.
mass_fluid_transport_process	environmental_interface	gas_path	OK	Sparging and exhaust conditioning must share a declared gas path.
sensing_instrumentation	control_compute_communication	sensor_bus	OK	DO, pH, mass and temperature measurements must feed process control.
control_compute_communication	mass_fluid_transport_process	pump_control	OK	Controller must command feed, harvest and gas-flow actuators.
safety_protection	structure_containment	pressure_relief	OK	Disposable culture containment must connect to pressure relief and sterile exhaust protection.

Interface Graph Summary

Module Nodes	Submodule Nodes	Shared Interface Edges	Required Edges	Missing Required Edges
9	27	39	6	0

Interface Contract Matrix

6/6 required contracts present. 15 shared interface families, 2 local-only declarations.

Interface	From Module	From Carrier	To Module	To Carrier	Status	Reason	Notes
sterile_fluid_path	Sterile Structure Containment	Single-use bag chamber; Sterile connector panel	Media Gas And Harvest Transport	Feed pump manifold; Harvest and sampling path; Sparger gas mixing	present	The sterile bag must connect to feed, harvest and sampling fluid paths.	Both endpoints expose the required interface through at least one submodule carrier.
sterile_boundary	Mixing And Agitation	Agitation drive train; Mixing impeller interface	Sterile Structure Containment	Single-use bag chamber; Sterile connector panel	present	Mixing drive must couple into the disposable culture boundary without breaking sterility.	Both endpoints expose the required interface through at least one submodule carrier.
gas_path	Media Gas And Harvest Transport	Sparger gas mixing	Thermal Gas Exhaust Interface	Exhaust filter train	present	Sparging and exhaust conditioning must share a declared gas path.	Both endpoints expose the required interface through at least one submodule carrier.
sensor_bus	Process Sensing Instrumentation	DO and pH sensing; Pressure, mass and temperature sensing; Gas flow sensing	Control Compute Communication	Process controller; Batch record gateway	present	DO, pH, mass and temperature measurements must feed process control.	Both endpoints expose the required interface through at least one submodule carrier.
pump_control	Control Compute Communication	Process controller	Media Gas And Harvest Transport	Feed pump manifold	present	Controller must command feed, harvest and gas-flow actuators.	Both endpoints expose the required interface through at least one submodule carrier.
pressure_relief	Aseptic And Pressure Safety	Sterile pressure safety	Sterile Structure Containment	Single-use bag chamber	present	Disposable culture containment must connect to pressure relief and sterile exhaust protection.	Both endpoints expose the required interface through at least one submodule carrier.

Local-only declarations to review later: mixing_drive (Mixing And Agitation); service_port (Control Compute Communication).

Interface Verification Gate

interface_review_ready

6 required interface row(s): 0 accepted, 6 review-ready, 0 pending evidence, 0 deferred, 0 blocked. Structural pass ratio 1; accepted evidence ratio 0.

Interface	From	To	Carrier Status	Verification Activity	Ledger	Status	Required Action
sterile_fluid_path	Sterile Structure Containment	Media Gas And Harvest Transport	both_endpoints_carried	interface_review:structure_containment:mass_fluid_transport_process:sterile_fluid_path	pending	review_ready	Collect named interface-review evidence through verification intake.
sterile_boundary	Mixing And Agitation	Sterile Structure Containment	both_endpoints_carried	interface_review:actuation_kinematics:structure_containment:sterile_boundary	pending	review_ready	Collect named interface-review evidence through verification intake.
gas_path	Media Gas And Harvest Transport	Thermal Gas Exhaust Interface	both_endpoints_carried	interface_review:mass_fluid_transport_process:environmental_interface:gas_path	pending	review_ready	Collect named interface-review evidence through verification intake.
sensor_bus	Process Sensing Instrumentation	Control Compute Communication	both_endpoints_carried	interface_review:sensing_instrumentation:control_compute_communication:sensor_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
pump_control	Control Compute Communication	Media Gas And Harvest Transport	both_endpoints_carried	interface_review:control_compute_communication:mass_fluid_transport_process:pump_control	pending	review_ready	Collect named interface-review evidence through verification intake.
pressure_relief	Aseptic And Pressure Safety	Sterile Structure Containment	both_endpoints_carried	interface_review:safety_protection:structure_containment:pressure_relief	pending	review_ready	Collect named interface-review evidence through verification intake.

Component Allocation Gate

allocation_ready

27/27 submodules ready; 0 review, 0 blocked. Component words 108; critical parts 11/11; duplicate groups 0. Allocation ratio 1; critical allocation ratio 1.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Sterile Structure Containment	Single-use bag chamber	ready	4	sterile_bound ary; sterile_fluid_p ath; mechanical_m ounts; service_acces s; pressure_relie f	single_use_bi oreactor_bag	none	Submodule has component candidates and no allocation blocker.
Sterile Structure Containment	Support frame and load path	ready	4	mechanical_m ounts; service_acces s	none	none	Submodule has component candidates and no allocation blocker.
Sterile Structure Containment	Sterile connector panel	ready	4	sterile_bound ary; sterile_fluid_p ath; service_acces s	sterile_tubing _and_connect or_set	none	Submodule has component candidates and no allocation blocker.
Mixing And Agitation	Agitation drive train	ready	4	mixing_drive; control_bus; sterile_bound ary	agitation_driv e	none	Submodule has component candidates and no allocation blocker.
Mixing And Agitation	Mixing impeller interface	ready	4	mixing_drive; sterile_bound ary; service_acces s	none	none	Submodule has component candidates and no allocation blocker.
Mixing And Agitation	Agitation safety feedback	ready	4	mixing_drive; sensor_bus; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Media Gas And Harvest Transport	Feed pump manifold	ready	4	sterile_fluid_p ath; pump_control; sensor_bus	peristaltic_fee d_pump	none	Submodule has component candidates and no allocation blocker.
Media Gas And Harvest Transport	Harvest and sampling path	ready	4	sterile_fluid_p ath; service_acces s; sensor_bus	none	none	Submodule has component candidates and no

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
							allocation blocker.
Media Gas And Harvest Transport	Sparger gas mixing	ready	4	gas_path; sterile_fluid_path; control_bus	sparger_and_gas_mix_manifold	none	Submodule has component candidates and no allocation blocker.
Thermal Gas Exhaust Interface	Temperature control loop	ready	4	thermal_loop; control_bus; sensor_bus	temperature_control_loop	none	Submodule has component candidates and no allocation blocker.
Thermal Gas Exhaust Interface	Exhaust filter train	ready	4	exhaust_path; gas_path; sterile_boundary; pressure_relief	exhaust_filter_and_pressure_relief_set	none	Submodule has component candidates and no allocation blocker.
Thermal Gas Exhaust Interface	Ambient enclosure conditions	ready	4	service_access; sensor_bus; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Process Sensing Instrumentation	DO and pH sensing	ready	4	sensor_bus; sterile_fluid_path	dissolved_oxygen_optical_sensor; single_use_ph_sensor	none	Submodule has component candidates and no allocation blocker.
Process Sensing Instrumentation	Pressure, mass and temperature sensing	ready	4	sensor_bus; thermal_loop; pressure_relief; mechanical_mounts	load_cell_set	none	Submodule has component candidates and no allocation blocker.
Process Sensing Instrumentation	Gas flow sensing	ready	4	sensor_bus; gas_path; exhaust_path	none	none	Submodule has component candidates and no allocation blocker.
Control Compute Communication	Process controller	ready	4	control_bus; sensor_bus; pump_control; alarm_bus	bioreactor_controller	none	Submodule has component candidates and no

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
							allocation blocker.
Control Compute Communication	Batch record gateway	ready	4	batch_record_network; service_port; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Control Compute Communication	Operator and alarm logic	ready	4	alarm_bus; control_bus; service_port	none	none	Submodule has component candidates and no allocation blocker.
Power Distribution	Control power panel	ready	4	mains_supply; protective_earth; control_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.
Power Distribution	Actuator power feeds	ready	4	mains_supply; control_bus; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Power Distribution	Backup power and shutdown	ready	4	mains_supply; alarm_bus; control_bus	none	none	Submodule has component candidates and no allocation blocker.
Aseptic And Pressure Safety	Sterile pressure safety	ready	4	sterile_boundary; pressure_relief; exhaust_path; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Aseptic And Pressure Safety	Aseptic setup controls	ready	4	sterile_boundary; service_access; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Aseptic And Pressure Safety	Operator electrical safety	ready	4	protective_earth; mains_supply; service_access	none	none	Submodule has component candidates and no

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
							allocation blocker.
Setup Calibration And Service	Single-use setup	ready	4	service_access; sterile_fluid_path; sterile_boundary	none	none	Submodule has component candidates and no allocation blocker.
Setup Calibration And Service	Sensor calibration access	ready	4	service_access; sensor_bus; sterile_fluid_path	none	none	Submodule has component candidates and no allocation blocker.
Setup Calibration And Service	Cleaning and turnaround	ready	4	service_access; batch_record_network	none	none	Submodule has component candidates and no allocation blocker.

Submodule Engineering Gate

submodule_engineering_review_required

0/27 submodules pass; 27 review, 0 blocked. Component words 108; linked requirements 4; carrier contracts 20/20; local-only interfaces 65. Review questions accepted 0/74; verification accepted 0/61; critical unpriced lines 11.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Sterile Structure Containment	Single-use bag chamber	review	4	5	1	3/3	0/5	0/6	1	0.6667	Review 2 local-only interface declaration(s): mechanical_mounts, service_access. 5 engineering review question(s) await acceptance. 5 verification activity row(s) await acceptance. 1 evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Sterile Structure Containment	Support frame and load path	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): mechanical_mounts, service_access. 2 engineering review question(s) await

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											acceptance. 1 verification activity row(s) await accepted evidence.
Sterile Structure Containment	Sterile connector panel	review	4	3	0	2/2	0/4	0/4	1	0.6667	Review 1 local-only interface declaration(s): service_access. 4 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Mixing And Agitation	Agitation drive train	review	4	3	1	1/1	0/3	0/4	1	0.6667	Review 2 local-only interface declaration(s): mixing_drive, control_bus. 3

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Mixing And Agitation	Mixing impeller interface	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): mixing_drive, service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Mixing And	Agitation safety	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Agitation	feedback										interface declaration(s): mixing_drive, sensor_bus, alarm_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Media Gas And Harvest Transport	Feed pump manifold	review	4	3	1	2/2	0/4	0/5	1	0.6667	Review 1 local-only interface declaration(s): sensor_bus. 4 engineering review question(s) await acceptance. 4 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufactur

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											er/MPN evidence.
Media Gas And Harvest Transport	Harvest and sampling path	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): service_access, sensor_bus. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Media Gas And Harvest Transport	Sparger gas mixing	review	4	3	0	2/2	0/4	0/4	1	0.6667	Review 1 local-only interface declaration(s): control_bus. 4 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											need source-backed cost/manufactur er/MPN evidence.
Thermal Gas Exhaust Interface	Temperature control loop	review	4	3	1	0/0	0/2	0/3	1	0.6667	Review 3 local-only interface declaration(s): thermal_loop, control_bus, sensor_bus. 2 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufactur er/MPN evidence.
Thermal Gas Exhaust Interface	Exhaust filter train	review	4	4	0	1/1	0/3	0/3	1	0.6667	Review 3 local-only interface declaration(s): exhaust_path, sterile_boundar

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											y, pressur e_relief. 3 engineer ing review questio n(s) await accepta nce. 2 verificat ion activity row(s) await accepte d evidenc e. 1 critical compon ent line(s) still need source-backed cost/man ufactur er/MPN evidenc e.

Thermal Gas Exhaust Interface	Ambient enclosure conditions	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): service_access, sensor_bus, alarm_bus. 2 engineer ing review questio n(s) await accepta nce. 1 verificat ion activity row(s) await accepte d
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Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											evidence.
Process Sensing Instrumentation	DO and pH sensing	review	4	2	0	1/1	0/3	0/3	2	0.6667	Review 1 local-only interface declaration(s): sterile_fluid_path. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await acceptance. evidence. 2 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Process Sensing Instrumentation	Pressure, mass and temperature sensing	review	4	4	0	1/1	0/3	0/3	1	0.6667	Review 3 local-only interface declaration(s): thermal_loop, pressure_relief, mechanical_mounts. 3 engineering review question(s) await

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											acceptance. 2 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Process Sensing Instrumentation	Gas flow sensing	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): gas_path, exhaust_path. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Control Computer Communication	Process controller	review	4	4	0	2/2	0/4	0/4	1	0.6667	Review 2 local-only interface declaration(s): control_bus, alarm_b

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											us. 4 engineer review question(s) await acceptance. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Control Compute Communication	Batch record gateway	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): batch_record_network, service_port. 3 engineer review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Control Computer Communication	Operator and alarm logic	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): alarm_bus, control_bus, service_port. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Power Distribution	Control power panel	review	4	4	0	0/0	0/2	0/1	0	0.6667	Review 4 local-only interface declaration(s): mains_supply, protective_earth, control_bus, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											evidence.
Power Distribution	Actuator power feeds	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): mains_supply, control_bus, protective_earth. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Power Distribution	Backup power and shutdown	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): mains_supply, alarm_bus, control_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											evidence.
Aseptic And Pressure Safety	Sterile pressure safety	review	4	4	0	1/1	0/3	0/2	0	0.6667	Review 3 local-only interface declaration(s): sterile_boundary, exhaust_path, alarm_bus. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Aseptic And Pressure Safety	Aseptic setup controls	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): sterile_boundary, service_access, alarm_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											evidence.
Aseptic And Pressure Safety	Operator electrical safety	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): protective_earth, mains_supply, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Setup Calibration And Service	Single-use setup	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): service_access, sterile_fluid_path, sterile_boundary. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											accepted evidence.
Setup Calibration And Service	Sensor calibration access	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): service_access, sensor_bus, sterile_fluid_path. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Setup Calibration And Service	Cleaning and turnaround	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): service_access, batch_record_network. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await acceptance

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
d evidence.											

Module Engineering Gate

module_engineering_review_required

0/9 modules pass; 9 review, 0 blocked. Submodules/components: 27/108. Interface carriers 12/12; review questions accepted 0/79; critical unpriced lines 11 across 6 module(s).

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Sterile Structure Containme nt	review	3	12	1	3/3	0/14	2	0.86	2 critical BoM line(s) need source- backed cost evidence. 14 engineerin g review question(s) await acceptanc e. 8 assumptio n row(s) need review or source evidence.
Mixing And Agitation	review	3	12	1	1/1	0/11	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 11 engineerin g review question(s) await acceptanc e. 6 assumptio n row(s) need review or source evidence.
Media Gas And Harvest Transport	review	3	12	1	3/3	0/14	2	0.86	2 critical BoM line(s) need source- backed cost evidence. 14 engineerin g review question(s) await acceptanc e. 9 assumptio n row(s) need review or

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
									source evidence.
Thermal Gas Exhaust Interface	review	3	12	1	1/1	0/12	2	0.86	2 critical BoM line(s) need source- backed cost evidence. 12 engineerin g review question(s) await acceptanc e. 8 assumptio n row(s) need review or source evidence.
Process Sensing Instrument ation	review	3	12	0	1/1	0/8	3	0.71	3 critical BoM line(s) need source- backed cost evidence. 8 engineerin g review question(s) await acceptanc e. 6 assumptio n row(s) need review or source evidence.
Control Compute Communic ation	review	3	12	0	2/2	0/7	1	0.71	1 critical BoM line(s) need source- backed cost evidence. 7 engineerin g review question(s) await acceptanc e. 3 assumptio n row(s) need

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
									review or source evidence.
Power Distributio n	review	3	12	0	0/0	0/4	0	0.71	4 engineerin g review question(s) await acceptanc e.
Aseptic And Pressure Safety	review	3	12	0	1/1	0/5	0	0.71	5 engineerin g review question(s) await acceptanc e. 2 assumptio n row(s) need review or source evidence.
Setup Calibration And Service	review	3	12	0	0/0	0/4	0	0.71	4 engineerin g review question(s) await acceptanc e. 2 assumptio n row(s) need review or source evidence.

Module Review Roll-Up

3 ready, 6 sourcing-blocked, 0 needing attention.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Sterile Structure Containment	sourcing	3	12	3/3	working_volume_l	single_use_bioreactor_bag; sterile_tubing_and_connectors_set	2 critical line(s) need source-backed evidence before BoM pass.
Mixing And Agitation	sourcing	3	12	1/1	working_volume_l	agitation_drive	1 critical line(s) need source-backed evidence before BoM pass.
Media Gas And Harvest Transport	sourcing	3	12	3/3	working_volume_l	peristaltic_feed_pump; sparger_and_gas_mix_manifold	2 critical line(s) need source-backed evidence before BoM pass.
Thermal Gas Exhaust Interface	sourcing	3	12	1/1	working_volume_l	temperature_control_loop; exhaust_filter_and_pressure_relief_set	2 critical line(s) need source-backed evidence before BoM pass.
Process Sensing Instrumentation	sourcing	3	12	1/1	none	dissolved_oxygen_optical_sensor; single_use_ph_sensor; load_cell_set	3 critical line(s) need source-backed evidence before BoM pass.
Control Compute Communication	sourcing	3	12	2/2	none	bioreactor_controller	1 critical line(s) need source-backed evidence before BoM pass.
Power Distribution	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.
Aseptic And Pressure Safety	ready	3	12	1/1	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Setup Calibration And Service	ready	3	12	0/0	none	none	No critical blockers; 12 candidate line(s) remain for later sourcing.

Engineering Review Pack

71 reviewer questions: 0 accepted, 42 ready, 29 needing review, 0 blocked. Coverage: 9 modules, 27 submodules, 6 interfaces, 2 calculations, 27 assumptions.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
assumption_resolution	major	needs review	EU-GMP-ANNEX-11	Can the EU-GMP-ANNEX-11 assumption be accepted for architecture review, or must it be replaced with evidence?	Batch record integrity, audit trail and access-control evidence.	none	none	none	none
assumption_resolution	major	needs review	IEC-61010	Can the IEC-61010 assumption be accepted for architecture review, or must it be replaced with evidence?	Electrical safety design, protective earth and abnormal operation evidence.	none	none	none	none
assumption_resolution	major	needs review	ISO-13485	Can the ISO-13485 assumption be accepted for architecture review, or must it be replaced with evidence?	Supplier quality evidence, change control and traceability where applicable.	none	none	none	none
assumption_resolution	major	needs review	USP-CLASS-VI	Can the USP-CLASS-VI assumption be accepted for architecture review, or must it be replaced with evidence?	Material certificates for single-use bag and tubing contact surfaces.	none	none	none	none
assumption_resolution	major	needs review	Agitation drive	Can the Agitation drive assumption be	Admit supplier, manufacturer, MPN where	actuation_kinematics	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				accepted for architecture review, or must it be replaced with evidence?	applicable, unit cost, lead time and evidence reference through sourcing intake.				
assumption_resolution	major	needs review	Bioreactor controller	Can the Bioreactor controller assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	control_compute_communication	none	none	none
assumption_resolution	major	needs review	Exhaust filter and pressure relief set	Can the Exhaust filter and pressure relief set assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	environmental_interface	none	none	none
assumption_resolution	major	needs review	Temperature control loop	Can the Temperature control loop assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	environmental_interface	none	none	none
assumption_resolution	major	needs review	Peristaltic feed pump	Can the Peristaltic feed pump assumption be accepted for	Admit supplier, manufacturer, MPN where applicable, unit cost,	mass_fluid_transport_process	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				architecture review, or must it be replaced with evidence?	lead time and evidence reference through sourcing intake.				
assumption_resolution	major	needs review	Sparger and gas-mix manifold	Can the Sparger and gas-mix manifold assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	mass_fluid_transport_process	none	none	none
assumption_resolution	major	needs review	Dissolved oxygen optical sensor	Can the Dissolved oxygen optical sensor assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	sensing_instrumentation	none	none	none
assumption_resolution	major	needs review	Single-use pH sensor	Can the Single-use pH sensor assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	sensing_instrumentation	none	none	none
assumption_resolution	major	needs review	Load cell set	Can the Load cell set assumption be accepted for architecture	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time	sensing_instrumentation	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				re review, or must it be replaced with evidence?	and evidence reference through sourcing intake.				
assumption_resolution	major	needs review	Single-use bioreactor bag	Can the Single-use bioreactor bag assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	structure_containment	none	none	none
assumption_resolution	major	needs review	Sterile tubing and connector set	Can the Sterile tubing and connector set assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	structure_containment	none	none	none
assumption_resolution	major	needs review	actuation_kinematics -> structure_containment	Can the actuation_kinematics -> structure_containment assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	sterile_boundary	none	none
assumption_resolution	major	needs review	control_compute_communication -> mass_fluid	Can the control_compute_communication -> mass_fluid	Review the interface carrier, direction, capacity,	none	pump_control	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
			_transport _process	_transport _process assumptio n be accepted for architectu re review, or must it be replaced with evidence?	safety margin and failure mode before detailed design or sourcing.				
assumptio n_resoluti on	major	needs review	mass_fluid _transport _process - > environme ntal_interf ace	Can the mass_fluid _transport _process - > environme ntal_interf ace assumptio n be accepted for architectu re review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	gas_path	none	none
assumptio n_resoluti on	major	needs review	safety_pro tection -> structure_ containme nt	Can the safety_pro tection -> structure_ containme nt assumptio n be accepted for architectu re review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	pressure_r elief	none	none
assumptio n_resoluti on	major	needs review	sensing_in strumenta tion -> control_co mpute_co mmunicati on	Can the sensing_in strumenta tion -> control_co mpute_co mmunicati on assumptio n be accepted for architectu	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed	none	sensor_bu s	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				re review, or must it be replaced with evidence?	design or sourcing.				
assumptio n_resoluti on	major	needs review	structure_ containme nt -> mass_fluid _transport _process	Can the structure_ containme nt -> mass_fluid _transport _process assumption be accepted for architectu re review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	sterile_flui d_path	none	none
assumptio n_resoluti on	major	needs review	Estimated CAPEX	Can the Estimated CAPEX assumption be accepted for architectu re review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumptio n_resoluti on	major	needs review	Annual culture volume	Can the Annual culture volume assumption be accepted for architectu re review, or must it be replaced with evidence?	Review the formula, input values and operating envelope before treating this metric as a performan ce claim.	none	none	none	none
assumptio n_resoluti on	major	needs review	Estimated annual OPEX	Can the Estimated annual OPEX assumption be accepted for	Admit source-backed BoM pricing before treating this cost	none	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				architectu re review, or must it be replaced with evidence?	metric as a claim.				
assumptio n_resoluti on	major	needs review	Aseptic And Pressure Safety	Can the Aseptic And Pressure Safety assumption be accepted for architectu re review, or must it be replaced with evidence?	Supply an engineerin g calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	structure_ containme nt; mass_fluid _transport _process; actuation_ kinematics ; environme ntal_interf ace	none	none	none
assumptio n_resoluti on	major	needs review	Sterile Process Closure	Can the Sterile Process Closure assumption be accepted for architectu re review, or must it be replaced with evidence?	Supply an engineerin g calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	structure_ containme nt; mass_fluid _transport _process; actuation_ kinematics ; environme ntal_interf ace	none	none	none
assumptio n_resoluti on	major	needs review	Single-Use Working Volume	Can the Single-Use Working Volume assumption be accepted for architectu re review, or must it be replaced with evidence?	Supply an engineerin g calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	structure_ containme nt; mass_fluid _transport _process; actuation_ kinematics ; environme ntal_interf ace	none	none	none
calculation _envelope	major	needs review	Bioreactor annual working volume	Does the Bioreactor annual working volume calculation support	Confirm batch duration, turnaroun d, cleaning/a ssembly	structure_ containme nt; mass_fluid _transport _process; actuation_	none	bioreactor _annual_w orking_vol ume_l	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				the architectu re envelope, or does the concept need revision?	time, hold steps and campaign model.	kinematics ; environme ntal_interf ace			
calculation _envelope	major	needs review	Bioreactor nominal bag volume	Does the Bioreactor nominal bag volume calculation support the architectu re envelope, or does the concept need revision?	Confirm bag platform, mixing turndown, foam allowance, sparging rate and working-volume range.	structure_ containme nt; mass_fluid _transport _process; actuation_ kinematics ; environme ntal_interf ace	none	bioreactor _nominal_ bag_volu me_l	none
interface_ contract	major	ready	Mixing And Agitation -> Sterile Structure Containme nt via sterile_bo undary	Is the sterile_bo undary interface between Mixing And Agitation and Sterile Structure Containme nt correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodul es, capacity limits and any deferred checks.	actuation_ kinematics ; structure_ containme nt	sterile_bo undary	none	Both endpoints expose the required interface through at least one submodul e carrier.
interface_ contract	major	ready	Control Compute Communication -> Media Gas And Harvest Transport via pump_con trol	Is the pump_con trol interface between Control Compute Communic ation and Media Gas And Harvest Transport correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodul es, capacity limits and any deferred checks.	control_co mpute_co mmunicati on; mass_fluid _transport _process	pump_con trol	none	Both endpoints expose the required interface through at least one submodul e carrier.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
interface_contract	major	ready	Media Gas And Harvest Transport -> Thermal Gas Exhaust Interface via gas_path	Is the gas_path interface between Media Gas And Harvest Transport and Thermal Gas Exhaust Interface correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	mass_fluid_transport_process; environmental_interface	gas_path	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Aseptic And Pressure Safety -> Sterile Structure Containment via pressure_relief	Is the pressure_relief interface between Aseptic And Pressure Safety and Sterile Structure Containment correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	safety_protection; structure_containment	pressure_relief	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Process Sensing Instrumentation -> Control Compute Communication via sensor_buses	Is the sensor_buses interface between Process Sensing Instrumentation and Control Compute Communication correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	sensing_instrumentation; control_compute_communication	sensor_buses	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Sterile Structure Containment -> Media Gas And Harvest Transport	Is the sterile_fluid_path interface between Sterile Structure Containment	Interface-review evidence naming both endpoint owners, carrier	structure_containment; mass_fluid_transport_process	sterile_fluid_path	none	Both endpoints expose the required interface through at least one

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculations	Blockers
			via sterile_fluid_path	nt and Media Gas And Harvest Transport correctly owned and physically carried?	submodules, capacity limits and any deferred checks.				submodule carrier.
module_allocation	major	ready	Mixing And Agitation	Does Mixing And Agitation have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	actuation_kinematics	mixing_drive; control_buses; sterile_boundary; sensor_buses; service_access; alarm_bus	bioreactor_working_volume; bioreactor_process_closure; bioreactor_aseptic_pressure_safety	none
module_allocation	major	ready	Control Compute Communication	Does Control Compute Communication have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	control_compute_communication	control_buses; sensor_buses; pump_control; batch_record_network; alarm_bus; service_port	none	none
module_allocation	major	ready	Thermal Gas Exhaust Interface	Does Thermal Gas Exhaust Interface have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	environmental_interface	thermal_loop; gas_path; exhaust_path; sterile_boundary; sensor_buses; control_buses; pressure_relief; service_access; alarm_bus	bioreactor_working_volume; bioreactor_process_closure; bioreactor_aseptic_pressure_safety	none
module_allocation	major	ready	Setup Calibration And Service	Does Setup Calibration And Service	Named engineering reviewer sign-off, change	maintenance_serviceability	service_access; sterile_fluid_path; sensor_buses	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				have the right responsibility boundary, submodule split and interface ownership for this concept?	request or deferral note against the module allocation.		s; batch_rec ord_netwo rk; sterile_bo undary		
module_all ocation	major	ready	Media Gas And Harvest Transport	Does Media Gas And Harvest Transport have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	mass_fluid _transport _process	sterile_flui d_path; gas_path; pump_con trol; control_bu s; sensor_bu s; service_ac cess	bioreactor _working_ volume; bioreactor _process_ closure; bioreactor _aseptic_ pressure_ safety	none
module_all ocation	major	ready	Power Distributio n	Does Power Distributio n have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	power_dis tribution	mains_sup ply; protective _earth; control_bu s; service_ac cess; alarm_bus	none	none
module_all ocation	major	ready	Aseptic And Pressure Safety	Does Aseptic And Pressure Safety have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	safety_pro tection	sterile_bo undary; pressure_r elief; exhaust_p ath; alarm_bus ; protective _earth; service_ac cess; mains_sup ply	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
module_all ocation	major	ready	Process Sensing Instrument ation	Does Process Sensing Instrument ation have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	sensing_in strumenta tion	sensor_bu s; sterile_flui d_path; gas_path; thermal_lo op; pressure_r elief; mechanica l_mounts; exhaust_p ath	none	none
module_all ocation	major	ready	Sterile Structure Containme nt	Does Sterile Structure Containme nt have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	structure_ containme nt	sterile_bo undary; sterile_flui d_path; mechanica l_mounts; service_ac cess; pressure_r elief	bioreactor _working_ volume; bioreactor _process_ closure; bioreactor _aseptic_ pressure_ safety	none
submodul e_allocation	major	ready	Mixing And Agitation / Agitation drive train	Do the componen t candidate s and interfaces assigned to Agitation drive train make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	actuation_ kinematics	mixing_dri ve; control_bu s; sterile_bo undary	bioreactor _working_ volume; bioreactor _process_ closure; bioreactor _aseptic_ pressure_ safety	none
submodul e_allocation	major	ready	Mixing And Agitation / Agitation safety feedback	Do the componen t candidate s and interfaces assigned to Agitation safety feedback make engineerin g sense for its	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	actuation_ kinematics	mixing_dri ve; sensor_bu s; alarm_bus	bioreactor _working_ volume; bioreactor _process_ closure; bioreactor _aseptic_ pressure_ safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				stated purpose?					
submodule_allocation	major	ready	Mixing And Agitation / Mixing impeller interface	Do the component candidates and interfaces assigned to Mixing impeller interface make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	actuation_kinematics	mixing_drive; sterile_boundary; service_access	bioreactor_working_volume; bioreactor_process_closure; bioreactor_aseptic_pressure_safety	none
submodule_allocation	major	ready	Control Compute Communication / Batch record gateway	Do the component candidates and interfaces assigned to Batch record gateway make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	control_compute_communication	batch_record_network; service_port; sensor_buses	none	none
submodule_allocation	major	ready	Control Compute Communication / Operator and alarm logic	Do the component candidates and interfaces assigned to Operator and alarm logic make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	control_compute_communication	alarm_buses; control_buses; service_port	none	none
submodule_allocation	major	ready	Control Compute Communication / Process controller	Do the component candidates and interfaces assigned to Process controller make engineering	Reviewer note confirming the submodule role, component allocation and any missing	control_compute_communication	control_buses; sensor_buses; pump_control; alarm_buses	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	ready	Thermal Gas Exhaust Interface / Ambient enclosure conditions	g sense for its stated purpose?	carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Ambient enclosure conditions make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	service_access; sensor_buses; alarm_bus	bioreactor_working_volume; bioreactor_process_closure; bioreactor_aseptic_pressure_safety	none
submodule_allocation	major	ready	Thermal Gas Exhaust Interface / Exhaust filter train	Do the component candidate s and interfaces assigned to Exhaust filter train make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	exhaust_path; gas_path; sterile_boundary; pressure_relief	bioreactor_working_volume; bioreactor_process_closure; bioreactor_aseptic_pressure_safety	none
submodule_allocation	major	ready	Thermal Gas Exhaust Interface / Temperature control loop	Do the component candidate s and interfaces assigned to Temperature control loop make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	thermal_loop; control_buses; sensor_buses	bioreactor_working_volume; bioreactor_process_closure; bioreactor_aseptic_pressure_safety	none
submodule_allocation	major	ready	Setup Calibration And Service / Cleaning and turnaround	Do the component candidate s and interfaces assigned to Cleaning	Reviewer note confirming the submodule role, component allocation	maintenance_serviceability	service_access; batch_record_network	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				and turnaround make engineering sense for its stated purpose?	and any missing carriers or capacity checks.				
submodule_allocation	major	ready	Setup Calibration And Service / Sensor calibration access	Do the component candidate s and interfaces assigned to Sensor calibration access make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	maintenance_service_ability	service_access; sensor_buses; sterile_fluid_path	none	none
submodule_allocation	major	ready	Setup Calibration And Service / Single-use setup	Do the component candidate s and interfaces assigned to Single-use setup make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	maintenance_service_ability	service_access; sterile_fluid_path; sterile_boundary	none	none
submodule_allocation	major	ready	Media Gas And Harvest Transport / Feed pump manifold	Do the component candidate s and interfaces assigned to Feed pump manifold make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	mass_fluid_transport_process	sterile_fluid_path; pump_control; sensor_buses	bioreactor_working_volume; bioreactor_process_closure; bioreactor_aseptic_pressure_safety	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodul e_allocation	major	ready	Media Gas And Harvest Transport / Harvest and sampling path	Do the componen t candidate s and interfaces assigned to Harvest and sampling path make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	mass_fluid _transport _process	sterile_flui d_path; service_ac cess; sensor_bu s	bioreactor _working_ volume; bioreactor _process_ closure; bioreactor _aseptic_ pressure_ safety	none
submodul e_allocation	major	ready	Media Gas And Harvest Transport / Sparger gas mixing	Do the componen t candidate s and interfaces assigned to Sparger gas mixing make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	mass_fluid _transport _process	gas_path; sterile_flui d_path; control_bu s	bioreactor _working_ volume; bioreactor _process_ closure; bioreactor _aseptic_ pressure_ safety	none
submodul e_allocation	major	ready	Power Distributio n / Actuator power feeds	Do the componen t candidate s and interfaces assigned to Actuator power feeds make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	power_dis tribution	mains_sup ply; control_bu s; protective _earth	none	none
submodul e_allocation	major	ready	Power Distributio n / Backup power and shutdown	Do the componen t candidate s and interfaces assigned to Backup power and shutdown make engineerin g sense	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or	power_dis tribution	mains_sup ply; alarm_bus ; control_bu s	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for its stated purpose?	capacity checks.				
submodule_allocation	major	ready	Power Distribution / Control power panel	Do the component candidate s and interfaces assigned to Control power panel make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	mains_supply; protective_earth; control_buses; service_access	none	none
submodule_allocation	major	ready	Aseptic And Pressure Safety / Aseptic setup controls	Do the component candidate s and interfaces assigned to Aseptic setup controls make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	safety_protection	sterile_boundary; service_access; alarm_bus	none	none
submodule_allocation	major	ready	Aseptic And Pressure Safety / Operator electrical safety	Do the component candidate s and interfaces assigned to Operator electrical safety make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	safety_protection	protective_earth; mains_supply; service_access	none	none
submodule_allocation	major	ready	Aseptic And Pressure Safety / Sterile pressure safety	Do the component candidate s and interfaces assigned to Sterile pressure	Reviewer note confirming the submodule role, component allocation	safety_protection	sterile_boundary; pressure_relief; exhaust_path; alarm_bus	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				safety make engineering sense for its stated purpose?	and any missing carriers or capacity checks.				
submodule_allocation	major	ready	Process Sensing Instrumentation / DO and pH sensing	Do the component candidates and interfaces assigned to DO and pH sensing make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	sensing_instrumentation	sensor_bu s; sterile_fluid_path	none	none
submodule_allocation	major	ready	Process Sensing Instrumentation / Gas flow sensing	Do the component candidates and interfaces assigned to Gas flow sensing make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	sensing_instrumentation	sensor_bu s; gas_path; exhaust_path	none	none
submodule_allocation	major	ready	Process Sensing Instrumentation / Pressure, mass and temperature sensing	Do the component candidates and interfaces assigned to Pressure, mass and temperature sensing make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	sensing_instrumentation	sensor_bu s; thermal_loop; pressure_relief; mechanical_mounts	none	none
submodule_allocation	major	ready	Sterile Structure Containment /	Do the component candidate	Reviewer note confirming the	structure_containment	sterile_boundary; sterile_fluid_path;	bioreactor_working_volume; bioreactor	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
			Single-use bag chamber	s and interfaces assigned to Single-use bag chamber make engineering sense for its stated purpose?	submodul e role, componen t allocation and any missing carriers or capacity checks.		mechanica l_mounts; service_ac cess; pressure_r elief	_process_ closure; bioreactor _aseptic_ pressure_ safety	
submodul e_allocatio n	major	ready	Sterile Structure Containme nt / Sterile connector panel	Do the componen t candidate s and interfaces assigned to Sterile connector panel make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	structure_ containme nt	sterile_bo undary; sterile_flui d_path; service_ac cess	bioreactor _working_ volume; bioreactor _process_ closure; bioreactor _aseptic_ pressure_ safety	none

Showing first 70 of 71 engineering review questions; export CSV for the full pack.

Engineering Sanity Checks

Check	Status	Value	Expected	Interpretation
Single-Use Working Volume	pass	50 L	1 to 2000 L for common single-use mammalian-cell systems; larger volumes need platform-specific bag and mixing evidence	50 L is plausible for a single-use mammalian-cell bioreactor before detailed mixing, oxygen-transfer and bag-platform evidence.
Sterile Process Closure	pass	present	sterile fluid path, gas path, sensor bus and pump control present	Media/feed/harvest paths, gas transfer, sensors and controller actions are connected before sourcing.
Aseptic And Pressure Safety	pass	present	sterile boundary, pressure relief, exhaust path, alarm bus and service access present	The architecture names the sterile and overpressure controls needed before BoM sourcing.

Engineering Calculation Ledger

2 calculations: 0 within deterministic envelope, 2 needing engineering review, 0 outside envelope, 0 blocked by missing inputs. These are arithmetic gates, not proof of real-world performance.

Calculation	Status	Formula	Result	Envelope	Interpretation	Evidence Required
Bioreactor nominal bag volume	needs review	$\text{working_volume_l} / 0.8$	62.5 L	80% fill fraction is a planning assumption before vendor bag geometry.	62.5 L nominal bag volume at 80% working fill.	Confirm bag platform, mixing turnaround, foam allowance, sparging rate and working-volume range.
Bioreactor annual working volume	needs review	$\text{working_volume_l} * 40$	2000 L/year	40 batches/year is a throughput assumption, not validated scheduling.	2000 L/year assumes 40 batches per year.	Confirm batch duration, turnaround, cleaning/assembly time, hold steps and campaign model.

Engineering Assumption Ledger

28 assumptions: 1 brief-supported, 6 model-present, 8 needing engineering review, 13 needing source evidence, 0 blocked. Architecture blockers 0; BoM blockers 13.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Working volume	brief_requirement	brief supported	no	no	Working volume = 50 L is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Annual culture volume	derived_metric	review required	no	no	Annual culture volume uses working_volume_l * batches_per_year with medium confidence.	Review the formula, input values and operating envelope before treating this metric as a performance claim.
Estimated CAPEX	derived_metric	source required	no	yes	Estimated CAPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Estimated annual OPEX	derived_metric	source required	no	yes	Estimated annual OPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Single-Use Working Volume	sanity_envelope	review required	no	no	Single-Use Working Volume currently evaluates as pass against 1 to 2000 L for common single-use mammalian-cell systems; larger volumes need platform-specific bag and mixing evidence.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Sterile Process Closure	sanity_envelope	review required	no	no	Sterile Process Closure currently evaluates as pass against sterile fluid path, gas path, sensor bus and pump control present.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Aseptic And Pressure Safety	sanity_envelope	review required	no	no	Aseptic And Pressure Safety currently evaluates as pass against sterile boundary, pressure relief,	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					exhaust path, alarm bus and service access present.	for this envelope.
structure_containment -> mass_fluid_transport_process	interface_closure	model present	no	no	sterile_fluid_path is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
actuation_kinematics -> structure_containment	interface_closure	model present	no	no	sterile_boundary is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
mass_fluid_transport_process -> environmental_interface	interface_closure	model present	no	no	gas_path is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
sensing_instrumentation -> control_compute_communication	interface_closure	model present	no	no	sensor_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
control_compute_communication -> mass_fluid_transport_process	interface_closure	model present	no	no	pump_control is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
safety_protection -> structure_containment	interface_closure	model present	no	no	pressure_relief is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Single-use bioreactor bag	critical_component	source required	no	yes	Single-use bioreactor bag is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Sterile tubing and connector set	critical_component	source required	no	yes	Sterile tubing and connector set is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Agitation drive	critical_component	source required	no	yes	Agitation drive is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Peristaltic feed pump	critical_component	source required	no	yes	Peristaltic feed pump is a critical architecture-derived component candidate with quantity 3 each.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Sparger and gas-mix manifold	critical_component	source required	no	yes	Sparger and gas-mix manifold is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Temperature control loop	critical_component	source required	no	yes	Temperature control loop is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Exhaust filter and pressure relief set	critical_component	source required	no	yes	Exhaust filter and pressure relief set is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Dissolved oxygen optical sensor	critical_component	source required	no	yes	Dissolved oxygen optical sensor is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Single-use pH sensor	critical_component	source required	no	yes	Single-use pH sensor is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Load cell set	critical_component	source required	no	yes	Load cell set is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Bioreactor controller	critical_component	source required	no	yes	Bioreactor controller is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
ISO-13485	compliance_evidence	review required	no	no	Quality management for medical-device-adjacent manufacturing systems is relevant to this product class in Global / supplier quality.	Supplier quality evidence, change control and traceability where applicable.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
USP-CLASS-VI	compliance_evidence	review required	no	no	Biocompatibility for polymer contact materials is relevant to this product class in US / global supplier evidence.	Material certificates for single-use bag and tubing contact surfaces.
IEC-61010	compliance_evidence	review required	no	no	Electrical safety for laboratory control equipment is relevant to this product class in UK/EU/global.	Electrical safety design, protective earth and abnormal operation evidence.
EU-GMP-ANNEX-11	compliance_evidence	review required	no	no	Computerised systems for GMP records is relevant to this product class in EU / pharma operations.	Batch record integrity, audit trail and access-control evidence.

Engineering Verification Plan

32 activities: 19 ready for review, 7 open, 6 blocked.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Sterile Structure Containment architecture allocation	ready	design_review	Sterile Structure Containment	working_volume_l	sterile_boundary; sterile_fluid_path; mechanical_mounts; service_access; pressure_relief	single_use_bioreactor_bag; bag_support_tray; port_clamp_set; bag_integrity_test_label; stainless_support_frame; load_cell_mounting_bracket_set; levelling_foot_set; lifting_eye_kit; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Sterile Structure Containment requirement calculations	ready	calculation	Sterile Structure Containment	working_volume_l	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Sterile Structure Containment critical components	blocked	source_evidence	Sterile Structure Containment	working_volume_l	none	single_use_bioreactor_bag; sterile_tubing_and_connector_set	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved	single_use_bioreactor_bag sterile_tubing_and_connector_set

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	
Review Mixing And Agitation architecture allocation	ready	design_review	Mixing And Agitation	working_volume_l	mixing_drive; control_bus; sterile_boundary; sensor_bus; service_access; alarm_bus	agitation_drive; magnetic_coupling_hub; drive_encoder; motor_mounting_plate; single_use_impeller_coupling; impeller_guard_insert; rocker_angle_stop; mixing_validation_coupling; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Mixing And Agitation requirement calculations	ready	calculation	Mixing And Agitation	working_volume_l	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Collect source-backed evidence for Mixing And Agitation critical components	blocked	source_evidence	Mixing And Agitation	working_volume_l	none	agitation_drive	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	agitation_drive
Review Media Gas And Harvest Transport architecture allocation	ready	design_review	Media Gas And Harvest Transport	working_volume_l	sterile_fluid_path; gas_path; pump_control; control_bus; sensor_bus; service_access	peristaltic_feed_pump; pump_tubing_cassette_set; media_bag_hanger; feed_line_pinch_clamp; harvest_transfer_tubing_set; single_use_sample_valve; waste_collection_bag; low_hold_up_connector; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Media Gas And Harvest Transport requirement calculations	ready	calculation	Media Gas And Harvest Transport	working_volume_l	none	none	Each linked requirement has a calculation, sanity check or clearly	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	
Collect source-backed evidence for Media Gas And Harvest Transport critical components	blocked	source_evidence	Media Gas And Harvest Transport	working_volume_l	none	peristaltic_feed_pump; sparger_and_gas_mix_manifold	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	peristaltic_feed_pump sparger_and_gas_mix_manifold
Review Thermal Gas Exhaust Interface architecture allocation	ready	design_review	Thermal Gas Exhaust Interface	working_volume_l	thermal_loop; gas_path; exhaust_path; sterile_boundary; sensor_bus; control_bus; pressure_relief; service_access; alarm_bus	temperature_control_loop; heater_blanket; cooling_plate; thermal_interface_mat; exhaust_filter_and_pressure_relief_set; condensate_knock_out_bottle; exhaust_heater_sleeve; pressure_relief_indicator	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						5; plus 4 more	No blocker or major architecture issue is attached to the module.	
Verify Thermal Gas Exhaust Interface requirement calculations	ready	calculation	Thermal Gas Exhaust Interface	working_volume_l	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Thermal Gas Exhaust Interface critical components	blocked	source_evidence	Thermal Gas Exhaust Interface	working_volume_l	none	temperature_control_loop; exhaust_filter_and_pressure_relief_set	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	temperature_control_loop exhaust_filter_and_pressure_relief_set
Review Process Sensing Instrumentation	ready	design_review	Process Sensing Instrumentation	none	sensor_bus; sterile_fluid_path; gas_path; thermal_loop;	dissolved_oxygen_optical_sensor; single_use_ph_sensor; sensor_package	Every submodule has a stated purpose and at least one	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
architecture allocation					pressure_relief; mechanical_mounts; exhaust_panth	h_reader; sensor_calibration_rec ord_card; load_cell_sen t; bag_pressu re_transduc er; culture_tem perature_pr obe; foam_detec tion_probe; plus 4 more	component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Collect source-backed evidence for Process Sensing Instrumentation critical components	blocked	source_evidence	Process Sensing Instrumentation	none	none	dissolved_oxygen_optical_sensor; single_use_ph_sensor; load_cell_sen t	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	dissolved_oxygen_optical_sensor single_use_ph_sensor load_cell_sen t
Review Control Compute Communication architecture allocation	ready	design_review	Control Compute Communication	none	control_bus; sensor_bus; pump_control; batch_record_network; alarm_bus; service_port	bioreactor_controller; i_o_terminal_slice_set; control_enclosure; recipe_execution_firmware; batch_historian_module; audit_trail_storage;	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						ethernet_switch; secure_time_source; plus 4 more	submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Collect source-backed evidence for Control Compute Communication critical components	blocked	source_evidence	Control Compute Communication	none	none	bioreactor_controller	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	bioreactor_controller
Review Power Distribution architecture allocation	ready	design_review	Power Distribution	none	mains_supply; protective_earth; control_bus; service_access; alarm_bus	rcd_mains_isolator; 24_v_dc_power_supply; terminal_block_rail; panel_earth_bar; pump_power_distribution_loom; agitation_drive_breaker; heater_output_relay; cable_gland_plate; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							architecture issue is attached to the module.	
Review Aseptic And Pressure Safety architecture allocation	ready	design_review	Aseptic And Pressure Safety	none	sterile_boundary; pressure_relief; exhaust_path; alarm_bus; protective_earth; service_access; mains_supply	single_use_pressure_relief_valve; vent_filter_integrity_tag; pressure_alarm_relay; bag_burst_shield; sterile_connection_checklist_card; connector_tamper_seal_set; pre_use_integrity_test_kit; operator_gowning_placard; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Review Setup Calibration And Service architecture allocation	ready	design_review	Setup Calibration And Service	none	service_access; sterile_fluid_path; sensor_bus; batch_record_network; sterile_boundary	bag_loading_fixture; tube_routing_template; connector_cap_organiser; setup_verification_barcode_sheet; ph_calibration_buffer_holder; do_zero_span_adapter; calibration_data_entry_hmi_page; sensor_replacement_guide_label; plus 4 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify sterile_fluid_path interface contract	ready	interface_review	Sterile Structure Containment -> Media Gas And	none	sterile_fluid_path	single_use_bioreactor_bag; bag_support_tray; port_clamp	Both endpoint modules declare the interface. At least one	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
			Harvest Transport			_set; bag_integrity_test_label; sterile_tubing_and_connector_set; aseptic_sampling_port; clamp_rail; connector_parking_bracket; plus 12 more	submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	
Verify sterile_boundary interface contract	ready	interface_review	Mixing And Agitation -> Sterile Structure Containment	none	sterile_boundary	agitation_drive; magnetic_coupling_hub; drive_encoder; motor_mounting_plate; single_use_impeller_coupling; impeller_guard_insert; rocker_angle_stop; mixing_validation_coupling; plus 8 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify gas_path interface contract	ready	interface_review	Media Gas And Harvest Transport -> Thermal Gas Exhaust Interface	none	gas_path	sparger_and_gas_mix_manifold; mass_flow_controller_bank; sterile_gas_filter_capsule; sparger_check_valve; exhaust_filter_and_pressure_relief_set; condensate_knock_out_bottle; exhaust_heater_sleeve; pressure_relief_indicator	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Verify sensor_bus interface contract	ready	interface_review	Process Sensing Instrumentation -> Control Compute Communication	none	sensor_bus	dissolved_oxygen_optical_sensor; single_use_ph_sensor; sensor_patch_reader; sensor_calibration_record_card; load_cell_set; bag_pressure_transducer; culture_temperature_probe; foam_detection_probe; plus 12 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify pump_control interface contract	ready	interface_review	Control Compute Communication -> Media Gas And Harvest Transport	none	pump_control	bioreactor_controller; i_o_terminal_slice_set; control_enclosure; recipe_execution_firmware; peristaltic_feed_pump; pump_tubing_cassette_set; media_bag_hanger; feed_line_pinch_clamp	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify pressure_relief interface contract	ready	interface_review	Aseptic And Pressure Safety -> Sterile Structure Containment	none	pressure_relief	single_use_pressure_relief_valve; vent_filter_integrity_tag; pressure_alarm_relay; bag_burst_shield; single_use_bioreactor_bag; bag_support_tray; port_clamp_set;	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						bag_integrity_test_label	the next detailed design pass.	
Plan evidence for ISO-13485	open	compliance_review	System Compliance	none	none	none	Supplier quality evidence, change control and traceability where applicable.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for USP-CLASS-VI	open	compliance_review	System Compliance	none	none	none	Material certificates for single-use bag and tubing contact surfaces.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for IEC-61010	open	compliance_review	System Compliance	none	none	none	Electrical safety design, protective earth and abnormal operation evidence.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for EU-GMP-ANNEX-11	open	compliance_review	System Compliance	none	none	none	Batch record integrity, audit trail and access-control evidence.	Compliance evidence not yet collected in scratch prototype.
Review risk control for Loss of sterility through connector or bag handling	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.
Review risk control for	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence	Risk evidence

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Cell culture loss from DO or pH control failure							and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	not yet reviewed in scratch prototype.
Review risk control for Disposable bag overpressure	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.

Verification Evidence Ledger

0 accepted, 0 rejected, 0 deferred, 26 pending, 0 blocked without evidence. Review coverage 0; acceptance ratio 0. Source-evidence activities remain in sourcing intake.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review Sterile Structure Containment architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Sterile Structure Containment requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Sterile Structure Containment critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Mixing And Agitation architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Mixing And Agitation requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Mixing And Agitation critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Media Gas And Harvest Transport architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Media Gas And Harvest Transport requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Media Gas And Harvest Transport critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Thermal Gas Exhaust Interface architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Thermal Gas Exhaust Interface requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Thermal Gas	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN,

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Exhaust Interface critical components					cost and lead-time evidence.
Review Process Sensing Instrumentation architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Process Sensing Instrumentation critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Control Compute Communication architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Control Compute Communication critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Power Distribution architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review Aseptic And Pressure Safety architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review Setup Calibration And Service architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify sterile_fluid_path interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify sterile_boundary interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify gas_path interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify sensor_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify pump_control interface contract	pending	interface_review			Collect reviewer evidence through

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Verify pressure_relief interface contract	pending	interface_review			the verification intake template.
					Collect reviewer evidence through the verification intake template.
					Collect reviewer evidence through the verification intake template.
Plan evidence for ISO-13485	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for USP-CLASS-VI	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for IEC-61010	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for EU-GMP-ANNEX-11	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Loss of sterility through connector or bag handling	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Cell culture loss from DO or pH control failure	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Disposable bag overpressure	pending	compliance_review			Collect reviewer evidence through the verification intake template.

Sterile Structure Containment

Supports a 50 L single-use culture volume while preserving the sterile disposable boundary.

- sterile_boundary
- sterile_fluid_path
- mechanical_mounts
- service_access
- pressure_relief

Single-use bag chamber

Carries the disposable culture bag and keeps ports accessible for sterile setup.

Components: Single-use bioreactor bag, bag support tray, port clamp set, bag integrity test label

Support frame and load path

Carries fluid mass, agitation loads and service handling forces.

Components: stainless support frame, load cell mounting bracket set, levelling foot set, lifting eye kit

Sterile connector panel

Organises aseptic feed, harvest, sampling and gas connections.

Components: Sterile tubing and connector set, aseptic sampling port, clamp rail, connector parking bracket

Mixing And Agitation

Transfers controllable mixing energy into the culture without compromising the disposable boundary.

mixing_drive	control_bus	sterile_boundary	sensor_bus
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Agitation drive train

Controls impeller or rocking motion for suspension culture.

Components: Agitation drive, magnetic coupling hub, drive encoder, motor mounting plate

Mixing impeller interface

Keeps the disposable impeller or rocker coupling aligned to the bag.

Components: single-use impeller coupling, impeller guard insert, rocker angle stop, mixing validation coupon

Agitation safety feedback

Detects stalled or excessive mixing conditions.

Components: motor current sensor, overspeed interlock input, drive fault relay, emergency drive stop contactor

Media Gas And Harvest Transport

Moves sterile media, feeds, harvest and process gases through controlled disposable paths.

sterile_fluid_path	gas_path	pump_control	control_bus	sensor_bus
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Feed pump manifold

Meters media, nutrient and base additions into the culture.

Components: Peristaltic feed pump, pump tubing cassette set, media bag hanger, feed line pinch clamp

Harvest and sampling path

Routes harvest and samples without opening the culture boundary.

Components: harvest transfer tubing set, single-use sample valve, waste collection bag, low-hold-up connector

Sparger gas mixing

Meters air, oxygen, nitrogen and CO2 into the disposable sparger path.

Components: Sparger and gas-mix manifold, mass-flow controller bank, sterile gas filter capsule, sparger check valve

Thermal Gas Exhaust Interface

Controls culture temperature and sterile exhaust conditions.

thermal_loop	gas_path	exhaust_path	sterile_boundary	sensor_bus
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Temperature control loop

Transfers heat into or out of the disposable culture vessel.

Components: Temperature control loop, heater blanket, cooling plate, thermal interface mat

Exhaust filter train

Maintains sterile exhaust and overpressure relief path.

Components: Exhaust filter and pressure relief set, condensate knock-out bottle, exhaust heater sleeve, pressure relief indicator

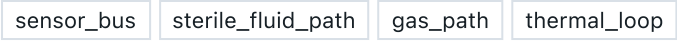
Ambient enclosure conditions

Manages local clean-bench or enclosure support conditions.

Components: ambient temperature probe, clean-zone status beacon, splash guard panel, wipe-down surface kit

Process Sensing Instrumentation

Measures culture and equipment state needed for closed-loop control and batch release evidence.



DO and pH sensing

Measures dissolved oxygen and pH through aseptic or single-use sensor paths.

Components: Dissolved oxygen optical sensor, Single-use pH sensor, sensor patch reader, sensor calibration record card

Pressure, mass and temperature sensing

Tracks working volume, pressure and thermal state.

Components: Load cell set, bag pressure transducer, culture temperature probe, foam detection probe

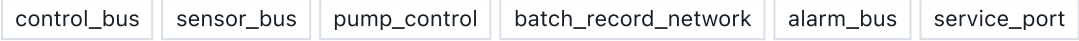
Gas flow sensing

Verifies inlet gas delivery and exhaust path condition.

Components: gas flow sensor set, exhaust pressure sensor, oxygen analyser cell, CO2 analyser cell

Control Compute Communication

Runs recipe, process control, alarms, historian and batch-record communication.



Process controller

Coordinates agitation, gas, pH, DO, temperature and pump loops.

Components: Bioreactor controller, I/O terminal slice set, control enclosure, recipe execution firmware

Batch record gateway

Exports process history and audit data.

Components: batch historian module, audit trail storage, ethernet switch, secure time source

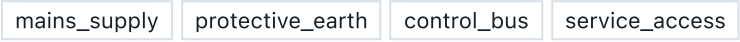
Operator and alarm logic

Presents alarms and controlled operator actions.

Components: alarm annunciator, operator HMI panel, role access key switch, batch acknowledge button

Power Distribution

Supplies protected mains and low-voltage power to pumps, controller, agitation and thermal hardware.



Control power panel

Distributes mains and DC control power to the bioreactor skid.

Components: RCD mains isolator, 24 V DC power supply, terminal block rail, panel earth bar

Actuator power feeds

Feeds agitation, pumps and thermal actuators.

Components: pump power distribution loom, agitation drive breaker, heater output relay, cable gland plate

Backup power and shutdown

Keeps control alive long enough for safe process stop.

Components: UPS buffer module, safe shutdown relay, power-fail input module, shutdown status lamp

Aseptic And Pressure Safety

Protects culture sterility, disposable pressure limits and operator exposure.

- sterile_boundary
- pressure_relief
- exhaust_path
- alarm_bus
- protective_earth

Sterile pressure safety

Prevents bag overpressure and sterile vent blockage.

Components: single-use pressure relief valve, vent filter integrity tag, pressure alarm relay, bag burst shield

Aseptic setup controls

Reduces setup contamination risk.

Components: sterile connection checklist card, connector tamper seal set, pre-use integrity test kit, operator gowning placard

Operator electrical safety

Keeps wet-process electrical risks controlled.

Components: emergency stop button, door interlock switch, leak tray float switch, protective earth test point

Setup Calibration And Service

Supports bag loading, sensor calibration, sterile turnaround and controlled service access.

- service_access
- sterile_fluid_path
- sensor_bus
- batch_record_network

Single-use setup

Guides bag installation and sterile connector management.

Components: bag loading fixture, tube routing template, connector cap organiser, setup verification barcode sheet

Sensor calibration access

Makes pre-batch sensor checks repeatable.

Components: pH calibration buffer holder, DO zero span adapter, calibration data entry HMI page, sensor replacement guide label

Cleaning and turnaround

Supports non-product-contact cleaning and post-batch reset.

Components: wipe-down kit holder, spill tray liner, spent bag removal cart interface, turnaround checklist card

Bill of Materials

Pricing status: architecture-derived candidate BoM only. Costs, suppliers, manufacturers and part numbers have not been independently sourced.

Total CAPEX BoM: £0

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
Single-use bioreactor bag	1	candidate	unpriced	unpriced				
bag support tray	1	candidate	unpriced	unpriced				
port clamp set	1	candidate	unpriced	unpriced				
bag integrity test label	1	candidate	unpriced	unpriced				
stainless support frame	1	candidate	unpriced	unpriced				
load cell mounting bracket set	1	candidate	unpriced	unpriced				
levelling foot set	1	candidate	unpriced	unpriced				
lifting eye kit	1	candidate	unpriced	unpriced				
Sterile tubing and connector set	1	candidate	unpriced	unpriced				
aseptic sampling port	1	candidate	unpriced	unpriced				
clamp rail	1	candidate	unpriced	unpriced				
connector parking bracket	1	candidate	unpriced	unpriced				
Agitation drive	1	candidate	unpriced	unpriced				
magnetic coupling hub	1	candidate	unpriced	unpriced				
drive encoder	1	candidate	unpriced	unpriced				
motor mounting plate	1	candidate	unpriced	unpriced				
single-use impeller coupling	1	candidate	unpriced	unpriced				
impeller guard insert	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
rocker angle stop	1	candidate	unpriced	unpriced				
mixing validation coupon	1	candidate	unpriced	unpriced				
motor current sensor	1	candidate	unpriced	unpriced				
overspeed interlock input	1	candidate	unpriced	unpriced				
drive fault relay	1	candidate	unpriced	unpriced				
emergency drive stop contactor	1	candidate	unpriced	unpriced				
Peristaltic feed pump	3	each	unpriced	unpriced				
pump tubing cassette set	1	candidate	unpriced	unpriced				
media bag hanger	1	candidate	unpriced	unpriced				
feed line pinch clamp	1	candidate	unpriced	unpriced				
harvest transfer tubing set	1	candidate	unpriced	unpriced				
single-use sample valve	1	candidate	unpriced	unpriced				
waste collection bag	1	candidate	unpriced	unpriced				
low-hold-up connector	1	candidate	unpriced	unpriced				
Sparger and gas-mix manifold	1	candidate	unpriced	unpriced				
mass-flow controller bank	1	candidate	unpriced	unpriced				
sterile gas filter capsule	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
sparger check valve	1	candidate	unpriced	unpriced				
Temperatur e control loop	1	candidate	unpriced	unpriced				
heater blanket	1	candidate	unpriced	unpriced				
cooling plate	1	candidate	unpriced	unpriced				
thermal interface mat	1	candidate	unpriced	unpriced				
Exhaust filter and pressure relief set	1	candidate	unpriced	unpriced				
condensate knock-out bottle	1	candidate	unpriced	unpriced				
exhaust heater sleeve	1	candidate	unpriced	unpriced				
pressure relief indicator	1	candidate	unpriced	unpriced				
ambient temperature probe	1	candidate	unpriced	unpriced				
clean-zone status beacon	1	candidate	unpriced	unpriced				
splash guard panel	1	candidate	unpriced	unpriced				
wipe-down surface kit	1	candidate	unpriced	unpriced				
Dissolved oxygen optical sensor	1	candidate	unpriced	unpriced				
Single-use pH sensor	1	candidate	unpriced	unpriced				
sensor patch reader	1	candidate	unpriced	unpriced				
sensor calibration record card	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
Load cell set	1	candidate	unpriced	unpriced				
bag pressure transducer	1	candidate	unpriced	unpriced				
culture temperature probe	1	candidate	unpriced	unpriced				
foam detection probe	1	candidate	unpriced	unpriced				
gas flow sensor set	1	candidate	unpriced	unpriced				
exhaust pressure sensor	1	candidate	unpriced	unpriced				
oxygen analyser cell	1	candidate	unpriced	unpriced				
CO2 analyser cell	1	candidate	unpriced	unpriced				
Bioreactor controller	1	candidate	unpriced	unpriced				
I/O terminal slice set	1	candidate	unpriced	unpriced				
control enclosure	1	candidate	unpriced	unpriced				
recipe execution firmware	1	candidate	unpriced	unpriced				
batch historian module	1	candidate	unpriced	unpriced				
audit trail storage	1	candidate	unpriced	unpriced				
ethernet switch	1	candidate	unpriced	unpriced				
secure time source	1	candidate	unpriced	unpriced				
alarm annunciator	1	candidate	unpriced	unpriced				
operator HMI panel	1	candidate	unpriced	unpriced				
role access key switch	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
batch acknowledge button	1	candidate	unpriced	unpriced				
RCD mains isolator	1	candidate	unpriced	unpriced				
24 V DC power supply	1	candidate	unpriced	unpriced				
terminal block rail	1	candidate	unpriced	unpriced				
panel earth bar	1	candidate	unpriced	unpriced				
pump power distribution loom	1	candidate	unpriced	unpriced				
agitation drive breaker	1	candidate	unpriced	unpriced				
heater output relay	1	candidate	unpriced	unpriced				
cable gland plate	1	candidate	unpriced	unpriced				
UPS buffer module	1	candidate	unpriced	unpriced				
safe shutdown relay	1	candidate	unpriced	unpriced				
power-fail input module	1	candidate	unpriced	unpriced				
shutdown status lamp	1	candidate	unpriced	unpriced				
single-use pressure relief valve	1	candidate	unpriced	unpriced				
vent filter integrity tag	1	candidate	unpriced	unpriced				
pressure alarm relay	1	candidate	unpriced	unpriced				
bag burst shield	1	candidate	unpriced	unpriced				
sterile connection checklist card	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
connector tamper seal set	1	candidate	unpriced	unpriced				
pre-use integrity test kit	1	candidate	unpriced	unpriced				
operator gowning placard	1	candidate	unpriced	unpriced				
emergency stop button	1	candidate	unpriced	unpriced				
door interlock switch	1	candidate	unpriced	unpriced				
leak tray float switch	1	candidate	unpriced	unpriced				
protective earth test point	1	candidate	unpriced	unpriced				
bag loading fixture	1	candidate	unpriced	unpriced				
tube routing template	1	candidate	unpriced	unpriced				
connector cap organiser	1	candidate	unpriced	unpriced				
setup verification barcode sheet	1	candidate	unpriced	unpriced				
pH calibration buffer holder	1	candidate	unpriced	unpriced				
DO zero span adapter	1	candidate	unpriced	unpriced				
calibration data entry HMI page	1	candidate	unpriced	unpriced				
sensor replacement guide label	1	candidate	unpriced	unpriced				
wipe-down kit holder	1	candidate	unpriced	unpriced				
spill tray liner	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
spent bag removal cart interface	1	candidate	unpriced	unpriced				
turnaround checklist card	1	candidate	unpriced	unpriced				

Cost And Sourcing

Sourcing Admission

Status	Candidate Lines	Admitted Prices	Unpriced Lines	Unpriced Critical Lines
not_started	108	0	108	11

Critical Sourcing Worklist

Component	Qty	Reason
Single-use bioreactor bag	1 candidate	Required class-critical line must be sourced before BoM can pass.
Sterile tubing and connector set	1 candidate	Required class-critical line must be sourced before BoM can pass.
Agitation drive	1 candidate	Required class-critical line must be sourced before BoM can pass.
Peristaltic feed pump	3 each	Required class-critical line must be sourced before BoM can pass.
Sparger and gas-mix manifold	1 candidate	Required class-critical line must be sourced before BoM can pass.
Temperature control loop	1 candidate	Required class-critical line must be sourced before BoM can pass.
Exhaust filter and pressure relief set	1 candidate	Required class-critical line must be sourced before BoM can pass.
Dissolved oxygen optical sensor	1 candidate	Required class-critical line must be sourced before BoM can pass.
Single-use pH sensor	1 candidate	Required class-critical line must be sourced before BoM can pass.
Load cell set	1 candidate	Required class-critical line must be sourced before BoM can pass.
Bioreactor controller	1 candidate	Required class-critical line must be sourced before BoM can pass.

No rejected source records.

BoM Sourcing Ledger

0/108 priced lines admitted. Critical coverage 0/11. Rejected evidence records: 0. Cost/manufacturer/MPN fields stay blank until evidence is admitted.

Component identity check: no duplicate componentWorldId groups detected.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
Single-use bioreactor bag	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
bag support tray	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
port clamp set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
bag integrity test label	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
stainless support frame	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
load cell mounting bracket set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
levelling foot set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
lifting eye kit	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Sterile tubing and connector set	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
aseptic sampling port	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
clamp rail	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
connector parking bracket	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
Agitation drive	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
magnetic coupling hub	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
drive encoder	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
motor mounting plate	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
single-use impeller coupling	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
impeller guard insert	candidate unpriced	unique	candidate	unpriced					Source after critical lines or

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									when detailed procurement planning starts.
rocker angle stop	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
mixing validation coupon	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
motor current sensor	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
overspeed interlock input	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
drive fault relay	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
emergency drive stop contactor	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Peristaltic feed pump	criticalunpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
pump tubing cassette set	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
media bag hanger	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
feed line pinch clamp	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
harvest transfer tubing set	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
single-use sample valve	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
waste collection bag	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
low-hold-up connector	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Sparger and gas-mix manifold	criticalunpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
mass-flow controller bank	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
sterile gas filter capsule	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
sparger check valve	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
Temperature control loop	criticalunpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
heater blanket	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
cooling plate	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
thermal interface mat	candidateunpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Showing first 40 of 108 ledger rows; export CSV for the full line-by-line ledger.

0 source-backed claims, 44 missing-source claims, 0 provenance violations. Critical missing-source claims: 44. Supplier, manufacturer, MPN and cost fields must appear here with source refs before procurement use.

Line	Claim Field	Status	Value	Source Ref	Next Action
Single-use bioreactor bag	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Single-use bioreactor bag	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Single-use bioreactor bag	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Single-use bioreactor bag	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Sterile tubing and connector set	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Sterile tubing and connector set	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Sterile tubing and connector set	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Sterile tubing and connector set	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Agitation drive	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Agitation drive	manufacturer	missing source			Admit source-backed

Line	Claim Field	Status	Value	Source Ref	Next Action
					supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Agitation drive	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Agitation drive	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Peristaltic feed pump	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Peristaltic feed pump	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Peristaltic feed pump	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Peristaltic feed pump	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Sparger and gas-mix manifold	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Sparger and gas-mix manifold	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Sparger and gas-mix manifold	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost

Line	Claim Field	Status	Value	Source Ref	Next Action
					evidence before BoM can pass.
Sparger and gas-mix manifold	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Temperature control loop	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Temperature control loop	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Temperature control loop	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Temperature control loop	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Exhaust filter and pressure relief set	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Exhaust filter and pressure relief set	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Exhaust filter and pressure relief set	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Exhaust filter and pressure relief set	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Line	Claim Field	Status	Value	Source Ref	Next Action
Dissolved oxygen optical sensor	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Dissolved oxygen optical sensor	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Dissolved oxygen optical sensor	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Dissolved oxygen optical sensor	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Single-use pH sensor	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Single-use pH sensor	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Single-use pH sensor	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Single-use pH sensor	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Load cell set	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Load cell set	manufacturer	missing source			Admit source-backed

Line	Claim Field	Status	Value	Source Ref	Next Action
					supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
Load cell set	mpn	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
Load cell set	unit_cost_gbp	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
Bioreactor controller	supplier	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
Bioreactor controller	manufacturer	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
Bioreactor controller	mpn	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.
Bioreactor controller	unit_cost_gbp	missing source			Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass.

Source Reference Quality Gate

No sourcing evidence references have been admitted. Manufacturer, MPN and cost fields must remain blank until source-backed evidence is admitted through sourcing intake.

BoM Evidence Trace Matrix

0/108 BoM row(s) are production-eligible; 0 protocol-only, 0 source-reference blocked, 11 critical unsourced. Priced-review rows 0; procurement use no; next row bom-trace:structure_containment_single_use_bag_chamber_single_use_bioreactor_bag.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
Single-use bioreactor bag	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
bag support tray	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
port clamp set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
bag integrity test label	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
stainless support frame	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
load cell mounting bracket set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
levelling foot set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
lifting eye kit	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Sterile tubing and connector set	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
aseptic sampling port	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
clamp rail	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
connector parking bracket	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Agitation drive	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
magnetic coupling hub	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
drive encoder	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
motor mounting plate	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
single-use impeller coupling	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
impeller guard insert	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
rocker angle stop	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
mixing validation coupon	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
motor current sensor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
overspeed interlock input	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
drive fault relay	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
emergency drive stop contactor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Peristaltic feed pump	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
pump tubing cassette set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
media bag hanger	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
feed line pinch clamp	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									critical sourcing is complete or procurement planning starts.
harvest transfer tubing set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
single-use sample valve	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
waste collection bag	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
low-hold-up connector	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Sparger and gas-mix manifold	critical unsource	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									BoM can pass.
mass-flow controller bank	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
sterile gas filter capsule	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
sparger check valve	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Temperature control loop	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
heater blanket	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
cooling plate	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									critical sourcing is complete or procurement planning starts.
thermal interface mat	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Exhaust filter and pressure relief set	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
condensate knock-out bottle	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
exhaust heater sleeve	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
pressure relief indicator	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
ambient temperature probe	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
clean-zone status beacon	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
splash guard panel	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
wipe-down surface kit	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
Dissolved oxygen optical sensor	critical unsource	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
Single-use pH sensor	critical unsource	critical	unpriced			0/4	no	no	Collect admissible source

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurem ent	Required Action
									evidence for this critical line before BoM can pass.

Showing first 50 of 108 BoM trace rows; export CSV for the full trace.

BoM Evidence Closure Plan

108 closure row(s): 11 ready, 0 blocked, 97 deferred. Actions: 11 collect source, 0 repair, 0 replace protocol, 97 candidate deferrals.
Procurement blockers 11; priced-review blockers 11; next row bom-close:actuation_kinematics_agitation_drive_train_agitation_drive.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
1	Agitation drive	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
2	Bioreactor controller	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
3	Exhaust filter and pressure relief set	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
4	Temperature control loop	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
5	Peristaltic feed pump	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
6	Sparger and gas-mix manifold	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
7	Dissolved oxygen optical sensor	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
8	Single-use pH sensor	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
9	Load cell set	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
10	Single-use bioreactor bag	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
11	Sterile tubing and connector set	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
12	drive encoder	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
13	magnetic coupling hub	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
14	motor mounting plate	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
15	drive fault relay	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
16	emergency drive stop contactor	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
17	motor current sensor	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
18	overspeed interlock input	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
19	impeller guard insert	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
20	mixing validation coupon	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
21	rocker angle stop	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
22	single-use impeller coupling	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
23	audit trail storage	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
24	batch historian module	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
25	ethernet switch	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
26	secure time source	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
27	alarm annunciator	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
28	batch acknowledged button	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
29	operator HMI panel	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
30	role access key switch	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
31	control enclosure	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
32	I/O terminal slice set	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
33	recipe execution firmware	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
34	ambient temperature probe	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
35	clean-zone status beacon	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
36	splash guard panel	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
37	wipe-down surface kit	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
38	condensate knock-out bottle	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
39	exhaust heater sleeve	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
40	pressure relief indicator	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
41	cooling plate	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
42	heater blanket	deferred	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
43	thermal interface mat	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
44	spent bag removal cart interface	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
45	spill tray liner	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
46	turnaroun d checklist card	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
47	wipe-down kit holder	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
48	calibration data entry HMI page	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
49	DO zero span adapter	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
50	pH calibration buffer holder	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 108 BoM closure rows; export CSV for the full queue.

Sourcing Batch Plan

2 batch(es): 1 active, 0 waiting, 1 deferred. Active rows 11; deferred rows 97. Critical source rows 11; repair rows 0; protocol replacements 0. Next batch sourcing-batch:2:critical_source_collection; next item critical_source_collection:1:agitation_drive.

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	Agitation drive	Agitation drive; Agitation drive manufacturer part number; Agitation drive catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Bioreactor controller	Bioreactor controller; Bioreactor controller manufacturer part number; Bioreactor controller catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Exhaust filter and pressure relief set	Exhaust filter and pressure relief set; Exhaust filter and pressure relief set manufacturer part number; Exhaust filter and pressure relief set catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Temperature control loop	Temperature control loop; Temperature control loop manufacturer part number; Temperature control loop catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Peristaltic feed pump	Peristaltic feed pump; Peristaltic feed pump manufacturer part	supplierName; manufacturer; mpn; unitCostGbp	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
					number; Peristaltic feed pump catalogue price GBP	p; evidence.ref		
2	Collect Critical Source Evidence	active	blocker	Sparger and gas-mix manifold	Sparger and gas-mix manifold; Sparger and gas-mix manifold manufacturer part number; Sparger and gas-mix manifold catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Dissolved oxygen optical sensor	Dissolved oxygen optical sensor; Dissolved oxygen optical sensor manufacturer part number; Dissolved oxygen optical sensor catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Single-use pH sensor	Single-use pH sensor; Single-use pH sensor manufacturer part number; Single-use pH sensor catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Load cell set	Load cell set; Load cell set manufacturer part number; Load cell set catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	Single-use bioreactor bag	Single-use bioreactor bag; Single-use bioreactor bag manufacturer part number; Single-use bioreactor bag catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Sterile tubing and connector set	Sterile tubing and connector set; Sterile tubing and connector set manufacturer part number; Sterile tubing and connector set catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
6	Defer Candidate-Only Sourcing	deferred	minor	drive encoder	drive encoder; drive encoder manufacturer part number; drive encoder catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	magnetic coupling hub	magnetic coupling hub; magnetic coupling hub manufacturer part number; magnetic coupling hub catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
6	Defer Candidate-Only Sourcing	deferred	minor	motor mounting plate	motor mounting plate; motor mounting plate manufacturer part number; motor mounting plate catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	drive fault relay	drive fault relay; drive fault relay manufacturer part number; drive fault relay catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	emergency drive stop contactor	emergency drive stop contactor; emergency drive stop contactor manufacturer part number; emergency drive stop contactor catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	motor current sensor	motor current sensor; motor current sensor manufacturer part number; motor current sensor catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	overspeed interlock input	overspeed interlock input; overspeed interlock input manufacturer part	no immediate evidence required unless this candidate is promoted to	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
					number; overspeed interlock input catalogue price GBP	procurement scope		
6	Defer Candidate- Only Sourcing	deferred	minor	impeller guard insert	impeller guard insert; impeller guard insert manufacturer part number; impeller guard insert catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate- Only Sourcing	deferred	minor	mixing validation coupon	mixing validation coupon; mixing validation coupon manufacturer part number; mixing validation coupon catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate- Only Sourcing	deferred	minor	rocker angle stop	rocker angle stop; rocker angle stop manufacturer part number; rocker angle stop catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate- Only Sourcing	deferred	minor	single-use impeller coupling	single-use impeller coupling; single-use impeller coupling manufacturer part number; single-use impeller coupling catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-	deferred	minor	audit trail storage	audit trail storage;	no immediate	no	candidate remains out

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
	Only Sourcing				audit trail storage manufacturer part number; audit trail storage catalogue price GBP	evidence required unless this candidate is promoted to procurement scope		of procurement totals

Procurement Readiness Gate

procurement_not_started

0/7 procurement areas pass; 2 review, 5 blocked. Critical production-eligible BoM rows 0/11; procurement-blocking closure rows 11; active sourcing rows 11; deferred sourcing rows 97. Source quality no_sourcing_evidence; evidence authenticity no_evidence; costing costing_not_started; BoM admission candidate_bom_review_required. Can use for procurement: no. Next action: Collect production-quality source evidence for critical BoM rows before procurement use..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture review	review	0.17	architecture_freeze_review_required; structurally ready yes, independent review accepted no.	none	Collect accepted engineering/interface/requirement review evidence before procurement use.
bom evidence trace	blocked	0	0/11 critical rows production-eligible; 11 critical unsourced, 0 protocol-only, 0 source-reference blocked.	structure_containment_single_use_bag_chamber_single_use_bioreactor_bag: Collect admissible source evidence for this critical line before BoM can pass. structure_containment_sterile_connector_panel_sterile_tubing_and_connectors_set: Collect admissible source evidence for this critical line before BoM can pass. actuation_kinematics_agitation_drive_train_agitation_drive: Collect admissible source evidence for this critical line before BoM can pass. mass_fluid_transport_process_feed_pump_manifold_peristaltic_feed_pump: Collect admissible source evidence for this critical line before BoM can pass. mass_fluid_transport_process_sparger_gas_mixing_sparger_and_gas_mix_manifold: Collect admissible source evidence for this critical line before BoM can pass. plus 6 more	Collect production-quality source evidence for critical BoM rows before procurement use.
sourcing batch closure	blocked	0.8981	11 procurement-blocking closure row(s); 11 active sourcing row(s), 97 deferred row(s).	bom-close:actuation_kinematics_agitation_drive_train_agitation_drive: Collect admissible source evidence for this critical line before BoM can pass. bom-close:control_compute_communication	Work the active sourcing batches until procurement-blocking closure rows are closed.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				_process_controller _bioreactor_controller: Collect admissible source evidence for this critical line before BoM can pass. bom-close:environmental_interface_exhaust_filter_train_exhaust_filter_and_pressure_relief_set: Collect admissible source evidence for this critical line before BoM can pass. bom-close:environmental_interface_temperature_control_loop_temperature_control_loop: Collect admissible source evidence for this critical line before BoM can pass. bom-close:mass_fluid_transport_process_feed_pump_manifold_peristaltic_feed_pump: Collect admissible source evidence for this critical line before BoM can pass. plus 6 more	
source reference quality	blocked	0	no_sourcing_evidence; 0/0 source-reference row(s) pass; 0 protocol, 0 placeholder.	none	Admit production source evidence before procurement readiness can start.
evidence authenticity	blocked	0	no_evidence; 0/0 evidence rows production-ready; 0 protocol.	No evidence records are present.	Add source and reviewer evidence before procurement readiness can start.
bom costing	blocked	0.14	costing_not_started ; critical priced 0/11; BoM total 0 GBP.	structure_containment_single_use_bag_chamber_single_use_bioreactor_bag: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. structure_containment_sterile_connector_panel_sterile_tubing_and_connector_set: Collect admissible supplier/manufactur	Close costing blockers before procurement use.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				er/MPN/unit-cost evidence before BoM can pass. actuation_kinematic s_agitation_drive_tr ain_agitation_drive: Collect admissible supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. mass_fluid_transpo rt_process_feed_pu mp_manifold_perist altic_feed_pump: Collect admissible supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. mass_fluid_transpo rt_process_sparger _gas_mixing_sparg er_and_gas_mix_m anifold: Collect admissible supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. plus 64 more	
prototype procurement policy	review	0	candidate_bom_revi ew_required; display mode candidate_only; prototype procurement use allowed no.	none	Prototype policy keeps procurement use disabled until external procurement review and approval are explicitly added.

Sourcing Evidence Pack

Evidence packets request source data only. They do not admit supplier, manufacturer, MPN or cost claims.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Single-use bioreactor bag	1 candidate	Single-use bioreactor bag; Single-use bioreactor bag manufacturer part number; Single-use bioreactor bag catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Sterile tubing and connector set	1 candidate	Sterile tubing and connector set; Sterile tubing and connector set manufacturer part number; Sterile tubing and connector set catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Agitation drive	1 candidate	Agitation drive; Agitation drive manufacturer part number; Agitation drive catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Peristaltic feed pump	3 each	Peristaltic feed pump; Peristaltic feed pump manufacturer part number; Peristaltic feed pump catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Sparger and gas-mix manifold	1 candidate	Sparger and gas-mix manifold; Sparger and gas-mix manifold manufacturer part number; Sparger and gas-mix manifold catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Temperature control loop	1 candidate	Temperature control loop; Temperature control loop manufacturer part number; Temperature control loop catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Exhaust filter and pressure relief set	1 candidate	Exhaust filter and pressure relief set; Exhaust filter and pressure relief set manufacturer part number; Exhaust filter and pressure relief set catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Dissolved oxygen optical sensor	1 candidate	Dissolved oxygen optical sensor; Dissolved oxygen optical sensor manufacturer part number; Dissolved oxygen optical sensor catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Single-use pH sensor	1 candidate	Single-use pH sensor; Single-use pH sensor manufacturer part number; Single-use pH sensor catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Load cell set	1 candidate	Load cell set; Load cell set manufacturer part number; Load cell set catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Bioreactor controller	1 candidate	Bioreactor controller; Bioreactor controller manufacturer part number; Bioreactor controller catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Cost model status: pending sourcing. Architecture is being reviewed before cost claims are admitted.

Cost Bucket	Estimate
CAPEX	£0
Annual OPEX	£0
NRE	£25,000

No admitted supplier spend yet. Supplier rows appear only after source-backed evidence records are admitted.

Regulatory And Risk

Standard	Title	Jurisdiction	Evidence Required
ISO-13485	Quality management for medical-device-adjacent manufacturing systems	Global / supplier quality	Supplier quality evidence, change control and traceability where applicable.
USP-CLASS-VI	Biocompatibility for polymer contact materials	US / global supplier evidence	Material certificates for single-use bag and tubing contact surfaces.
IEC-61010	Electrical safety for laboratory control equipment	UK/EU/global	Electrical safety design, protective earth and abnormal operation evidence.
EU-GMP-ANNEX-11	Computerised systems for GMP records	EU / pharma operations	Batch record integrity, audit trail and access-control evidence.

Hazard	S	O	D	RPN	Mitigation
Loss of sterility through connector or bag handling	9	4	5	180	Closed sterile connectors, pre-use integrity checks and operator setup checklist.
Cell culture loss from DO or pH control failure	8	4	4	128	Redundant alarming, sensor calibration and conservative gas/feed control limits.
Disposable bag overpressure	8	3	3	72	Pressure relief, vent filter monitoring and pump/gas interlocks.

Section Issues And Score

bom

- blocker / critical_part_unpriced: Single-use bioreactor bag is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Sterile tubing and connector set is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Agitation drive is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Peristaltic feed pump is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Sparger and gas-mix manifold is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Temperature control loop is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Exhaust filter and pressure relief set is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Dissolved oxygen optical sensor is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Single-use pH sensor is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Load cell set is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Bioreactor controller is critical but has no credible unit cost.
- major / low_priced_line_ratio: Only 0% of BoM lines have prices.

Section Score Ledger

Rule: start at 9, subtract 3 per blocker, 1.5 per major, 0.5 per minor, floor at 1. Mean score 7.67. This is a deterministic gate score from current validators, not an external reviewer score or proof of real-world engineering quality. Reviewer acceptance is tracked separately in the verification ledger.

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
executive_summary	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
brief_requirements	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
design_modules	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
bom	9	11/1/0	-34.5	1	below target	blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); major/low_priced_line_ratio (-1.5)	Score is base minus deterministic deductions for attached section issues.

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
feasibility_notes	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
sources_references	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
Mean	7.67						